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A new interpretable fault diagnosis method based on belief rule base and probability table



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Abstract It is vital to establish an interpretable fault diagnosis model for critical equipment. Belief Rule Base (BRB) is an interpretable expert system gradually applied in fault diagnosis. However, the expert knowledge cannot be utilized to establish the initial BRB accurately if there are multiple referential grades in different fault features. In addition, the interpretability of BRB-based fault diagnosis is destroyed in the optimization process, which reflects in two aspects: deviation from the initial expert judgment and over-optimization of parameters. To solve these problems, a new interpretable fault diagnosis model based on BRB and probability table, called the BRB-P, is proposed in this paper. Compared with the traditional BRB, the BRB-P constructed by the probability table is more accurate. Then, the interpretability constraints, i.e., the credibility of expert knowledge, the penalty factor and the rule-activation factor, are inserted into the projection covariance matrix adaption evolution strategy to maintain the interpretability of BRB-P. A case study of the aerospace relay is conducted to verify the effectiveness of the proposed method.
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1. Introduction

Critical equipment, such as the electro-hydraulic servo and the aerospace relay, plays a significant role in the operation of complex systems. Once the fault occurs, if unable to be ruled out timely, it may cause considerable economic losses. Fault diagnosis is an effective measure to diagnose abnormal working states.

According to different information used in fault diagnosis, three types of methods are usually adopted, including the model-based method,[1–2] the data-based method[3] and the expertise-based method.[4] In the model-based methods, the fault diagnosis model is established by analyzing the fault mechanism of equipment.[5] Good transparency, precise mechanism and strong interpretability constitute its main advantages. Nonetheless, this method depends on establishing an accurate fault mechanism model. If there are a large number of fault samples, a data-based method such as the neural net-

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work can be employed to describe the causal relationship between fault features and fault modes.[6–9] The model established in this method works in black-box perspective and is entirely independent from system mechanism.[10] When there is a lack of fault samples, and the system mechanism model is challenging to obtain, the expert knowledge and historical experience can be utilized to construct the fault model, which is the expertise-based method.[11] However, the diagnostic accuracy is relevant to the expert knowledge due to the subjective deviation of experts.

Based on the above analysis, it is necessary to establish an excellent fault diagnosis model for critical equipment that generally has the following characteristics:

(1) Considering the complex structure, the accurate mechanism model is hard to be established.
(2) In most cases, the equipment works in the normal state with limited fault samples, and the sample characteristic tends to be "low density".[12]
(3) Due to the particularity, there is a demand to establish an understandable fault diagnosis model for critical equipment, which increases the trust between humans and the model.[13]

Based on the above characteristics, this paper aims to establish an interpretable fault diagnosis model for critical equipment. Interpretability in this paper means that the operation of a model can be understood by humans.[14–15]

Yang et al.[16] proposed the Belief Rule Base (BRB) inference methodology using the evidential reasoning approach, which is a typical expert system. The rule of BRB has the understandable knowledge form of "IF-THEN".[17] BRB is developed based on the evidence theory.[18] The uncertainty can be measured by belief function rather than probability in evidence theory. There is no need for a large number of historical samples to determine parameters such as the prior distribution. The quantitative information and qualitative knowledge can be effectively combined to construct the model. The evidence theory has been applied in information fusion, expert system and decision analysis based on the above advantages. There is a general knowledge representation scheme based on the belief structure for various types of inputs, such as quantitative data and qualitative information. Thus, BRB has the ability of complex nonlinear modeling based on semantic information.[19] When the fault samples are limited, or the mechanism model is difficult to be obtained, the BRB can be established based on the expert knowledge or historical experience. The parameters of BRB can be modified during the optimization process. The reasoning engine of BRB is the Evidential Reasoning (ER) approach that is used to fuse the activated rules. The ER approach is a generalization form of the traditional Bayesian inference rule.[20] BRB has been widely employed in various fields, such as disease diagnosis, engineering manufacturing, industrial management,[21–25] etc.

However, there are still two problems in the BRB-based fault diagnosis.

(1) In the modeling process, many fault features and modes exist in the system mentioned above. The ability of experts to deal with complex systems is limited. The expert knowledge cannot be accurately embedded in BRB when there are a large number of rules.
(2) In the optimization process, the interpretability of BRB may be destroyed, and the trust degree between humans and the model is reduced. On the one hand, the initial BRB is given and effectively explained by experts. Inevitably, the optimized BRB may deviate from the initial expert judgment so that the optimized rules cannot be understood by experts. On the other hand, the parameters of BRB are over-optimized.[26] In other words, all parameters of a fault diagnosis model are optimized previously whether they work or not during the training process.

To accurately establish the model of complex systems, Chang et al.[17] proposed a disjunctive belief rule base modeling method, in which BRB model was constructed under the disjunctive assumption rather than the conjunctive assumption. A new approximate BRB model, named ABRB, was proposed by Cao et al.[27] The input of a rule is converted into a single input form by introducing the independency factor in ABRB. Reducing the number of rules in these methods is convenient to embed expert knowledge. In addition, Zhang et al.[28] proposed a method of automatically generating initial parameters for BRB by means of cloud model and partial rules given by experts. The establishment of BRB has been conducted with in-depth research, but the interpretability of the model has not been mentioned in the above literature. With the attention paid to the interpretability of algorithms, the research on the interpretability of BRB has made some progress. Yang et al.[29] put forward a viewpoint that the parameters of BRB should be fine-tuned in optimization. Zhou et al.[19] stated the intrinsic interpretability of BRB and proposed a new health-state assessment model based on BRB with interpretability.[30] Cao et al.[26] summarized the interpretability of BRB and proposed a modified optimization algorithm with the interpretability constraints. The interpretability of BRB was defined, and the evaluation indicator of interpretability was proposed by You et al.[31]

In this paper, a new fault diagnosis model, called BRB-P, is constructed based on BRB with a probability table and three interpretability constraints. It considers both the establishment of BRB and the maintenance of model interpretability. The expert knowledge is expressed as the probability table to establish the initial fault diagnosis model. In this way, the number of initialized parameters is decreased to be convenient for applying the expert knowledge. Meanwhile, the completeness of the rule base can be preserved. Then, the parameters of the initial BRB-P are optimized by the limited fault samples to reduce the error brought by the subjectivity of experts. In this paper, the Projection Covariance Matrix Adaption Evolution Strategy (P-CMA-ES) optimization algorithm is utilized to optimize the parameters of BRB-P, and interpretability constraints are added to maintain the interpretability of the fault diagnosis model. Specifically, there are three interpretability constraints, introducing the credibility of expert knowledge, the penalty factor and rule-activation factor to the optimization algorithm, respectively. The credibility of expert knowledge and the penalty factor can avoid deviating from the initial judgment of experts in optimization. The problem of parameters over-optimization can be solved by introducing the rule-activation factor.

The remainder of this paper is organized as follows. In Section 2, the relevant problems of BRB for fault diagnosis are

discussed, and the framework of BRB-P is briefly introduced. In Section 3, the establishment and reasoning method of BRB-P is proposed. The optimization of BRB-P with interpretability constraints is conducted in Section 4. A case study of the relay fault diagnosis is performed to verify the effectiveness of the proposed method in Section 5. This paper is concluded in Section 6.

2. Problem formulation and framework of BRB-P model

In this section, the problem formulation of fault diagnosis based on BRB is presented. Then, the basic framework of BRB-P is constructed.

2.1. Problem formulation of fault diagnosis based on BRB

Suppose there are M fault features and N fault modes in the system. The BRB is utilized to describe the relationship between them. A rule base is composed of a series of rules, which are the comprehensible form of "IF-THEN". The structure of the k th rule is expressed as follows:

$$\begin{gathered} R_k: \text{ If } (x_1^k \text{ is } A_1^k) \wedge (x_2^k \text{ is } A_2^k) \wedge \cdots \wedge (x_M^k \text{ is } A_M^k), \\ \text{then } \{(D_1, \widehat{\beta}_{1,k}), (D_2, \widehat{\beta}_{2,k}), \cdots, (D_N, \widehat{\beta}_{N,k})\} \\ \text{with the rule weight } \theta_k \text{ and fault feature weights } \delta_1, \delta_2, \cdots, \delta_M \end{gathered} \tag{1}$$

where $\wedge$ is the logical conjunction representing "AND"; x_i^k and D_j are the i th fault feature in R_k and the j th fault mode respectively; A_i^k denotes the referential value of the i th fault feature in R_k; $\hat{\beta}_{j,k}$ is the belief degree of the j th fault mode in R_k; θ_k and δ_i indicate the weights of R_k and the i th fault feature, which represent the relative importance to other rules and fault features respectively. The initial values of $\hat{\beta}_{j,k}$, θ_k and δ_i should be given by experts to ensure the interpretability of BRB model. Then, the parameters of BRB are optimized by using an optimization algorithm according to the limited fault samples. Nevertheless, there are two problems in the process of utilizing the BRB to diagnose faults. The specific analysis is made as follows:

Problem 1. All the referential values of fault features should be traversed when establishing a BRB-based fault diagnosis model. Thus, the number of rules in BRB can be calculated by

$$K = \prod_{m=1}^{M} c_m \tag{2}$$

where c_m is the number of referential grades of the m th fault feature. As can be seen in Eq. (2), K increases exponentially with M. The initialized parameters given by experts include the belief degrees, fault feature weights and rule weights. The total number of initialized parameters can be calculated by

$$\mathrm{Num_p} = N \times K + M + K \tag{3}$$

Taking the fault diagnosis of the relay as an example, 4 fault features and 6 fault modes are the input and output variables of the fault diagnosis model. 3 referential grades are assigned for each fault feature. The k th rule is shown in Eq. (4). There are 81 ($3^4 = 81$) rules in BRB and 571 initialized parameters in the traditional BRB, which should be given by experts to maintain the interpretability of BRB. However, the ability to process a great deal of information for experts is limited.[32–33] Zhang et al.[28] pointed out that experts were hard to apply knowledge and historical experience to construct the BRB-based model when there were a large number of parameters. In other words, BRB cannot be accurately established by experts. It is necessary to propose a new mechanism to transform the expert knowledge into the initial BRB.

$$\begin{gathered} R_k: \text{ If } (x_1 \text{ is } A_1^k) \wedge (x_2 \text{ is } A_2^k) \wedge (x_3 \text{ is } A_3^k) \wedge (x_4 \text{ is } A_4^k), \\ \text{then } \{(D_1, \widehat{\beta}_{1,k})\ (D_2, \widehat{\beta}_{2,k}), \cdots, (D_6, \widehat{\beta}_{6,k})\} \\ \text{with the rule weight } \theta_k \text{ and fault feature weights } \delta_1, \delta_2, \delta_3, \delta_4 \end{gathered} \tag{4}$$

Problem 2. After the initial BRB is established by experts, the optimization process is employed to reduce the model error brought by the subjectivity of experts.[34] The target of the current optimization is to improve the accuracy of the model while maintaining interpretability is ignored. In this process, the interpretability of BRB is destroyed in two aspects: the deviation from the initial judgment of experts and the over-optimization of parameters. The initial judgment of experts is a key element of the interpretability of BRB. In other words, the model result is consistent with the experts' cognition. In this paper, we assume that the experts are authoritative. It is noted that the target of optimization should be to fine-tune the parameters of the BRB model based on the initial judgment. However, the optimized rules may be different from the initial rules to a large extent, which means that the deviation from the expert knowledge occurs.[35–36] The final result shows that experts are unable to understand the optimized BRB. For instance, Liu et al.[37] established the BRB model for fault diagnosis of an airborne missile, whose partial rules are shown in Table 1. Refrigeration noise voltage (x_1) and refrigerating flow (x_2) are selected as fault features. Taking the 8th rule as an example, the belief degrees given by experts are (0.8600, 0.0000, 0.1400). Fault D_1 occurs when the input is exactly the reference values of fault features in the 8th rule. However, the corresponding result of the optimized BRB indicates that the fault D_3 occurs in the same case. It is noted that the optimized rules are inconsistent with the experts' cognition.

In addition, the over-optimization problem of parameters means that the parameters of unactivated rules are optimized, which weakens the interpretability of BRB.[30] The existing normal or fault states are included in the fault diagnosis model. However, the limited samples may contain the partial states of a system. In other words, some rules unactivated in the training process never play roles in the fusion process. It is irrational that the parameters of a fault diagnosis model are totally optimized previously whether they work or not.

2.2. Framework of fault diagnosis model based on BRB-P

To solve the problems mentioned above, a new fault diagnosis model based on BRB-P is proposed, whose framework is depicted in Fig. 1. The following part highlights the functions of each module in Fig. 1.

In the modeling process, the initial BRB-P is established by the probability table rather than being given directly by experts

Table 1 Partial rules of BRB in fault diagnosis model of an airborne missile.

No.	x_1	x_2	BRB	$\{(D_1,\hat{\beta}_{1,k}),(D_2,\hat{\beta}_{2,k}),(D_3,\hat{\beta}_{3,k})\}$
2	Normal	Small	Initial	{(Detector fault,0.0000), (Cooler fault,0.1900), (Nitrogen pipeline fault,0.8100)}
			Optimized	{(Detector fault,0.0455), (Cooler fault,0.7898), (Nitrogen pipeline fault,0.1647)}
5	Large	Small	Initial	{(Detector fault,0.0000), (Cooler fault,1.0000), (Nitrogen pipeline fault,0.0000)}
			Optimized	{(Detector fault,0.2733), (Cooler fault,0.4729),(Nitrogen pipeline fault,0.2538)}
8	Very large	Small	Initial	{(Detector fault,0.8600), (Cooler fault,0.0000), (Nitrogen pipeline fault,0.1400)}
			Optimized	{(Detector fault,0.0598), (Cooler fault,0.0834), (Nitrogen pipeline fault,0.8568)}

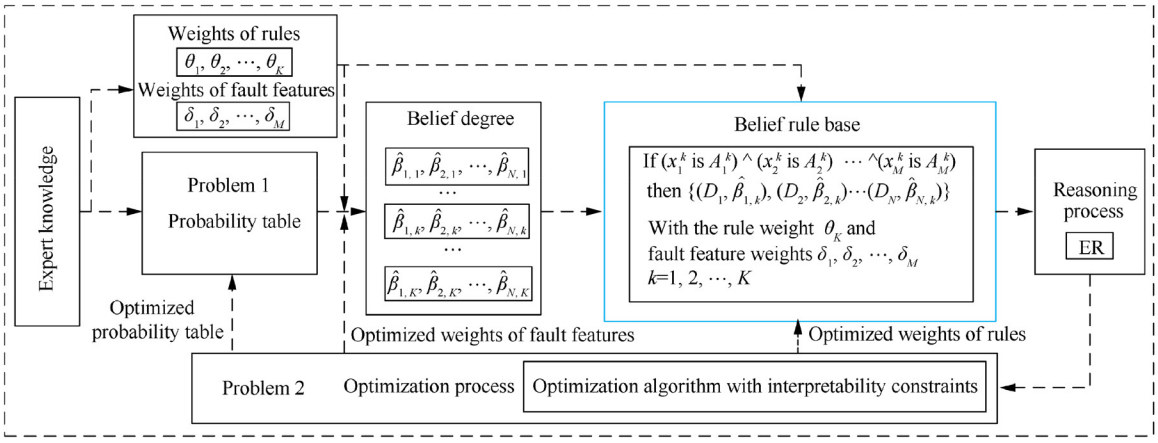


Fig. 1 Interpretable fault diagnosis model based on BRB-P.

to solve Problem 1. The factors of a probability table are produced by the expert knowledge, which decreases the number of initialized parameters and preserves the completeness of BRB.

The reasoning process of BRB-P includes the transformation of input information, calculation of rule activation weights and rule fusion by ER.[38] In this process, the mapping relationship between fault features ($\boldsymbol{x}$) and fault modes (D) is presented as follows:

$$S(\boldsymbol{x}) = \{(D_j, \beta_j),\ j = 1, 2, \cdots, N\} \tag{5}$$

It is necessary to utilize the existing fault samples to improve the model's performance since the parameters may be subjectively given by experts. The P-CMA-ES optimization algorithm is introduced to enhance the accuracy of the initial BRB-P. In allusion to Problem 2, interpretability constraints are inserted into P-CMA-ES to maintain the interpretability of BRB.[12–13] Both the deviation from the initial judgment of experts and the over-optimization problem of parameters are avoided.

The detailed fault diagnosis process is presented in Sections 3 and 4.

3. Establishment and reasoning of BRB-P model

In this section, BRB-P is exploited to describe the mapping relationship between fault features and fault modes. The probability table is introduced to establish the initial BRB-P model, and the ER approach is used to fuse the activated rules to generate diagnostic results.

3.1. Establishment of BRB-P model

In this subsection, the transition process of expert knowledge to the initial BRB-P is conducted by a probability table. The factors of the probability table are transformed into belief degrees of the initial BRB-P.

(1) Generation of probability table

The expert knowledge is aggregated into a probability table. The factors of the probability table are the values of probability. The implication of each factor is the probability of fault feature near a specific referential value when a fault mode occurs. For instance, the probability of Pull-in Voltage (PV) near referential grade L is 0.1 when fault contact wear (D_1) occurs in the fault diagnosis of a relay. The specific meaning can be expressed as

$$P(\mathrm{PV} = L | D_1 \text{ happened}) = 0.1 \tag{6}$$

A more general form is presented as

$$P(x_m = A_m^i | D_n \text{ happened}) = P_{m,i}^n, \sum_{i=1}^{c_m} P_{m,i}^n = 1\ (0 \leqslant P_{m,i}^n \leqslant 1) \tag{7}$$

where c_m is the number of referential grades of x_m. The complete probability table is shown in Table 2.

Remark: The number of initialized parameters in BRB-P is less than that in the traditional BRB. The number of initialized parameters in the traditional BRB is expressed as

$$\mathrm{Num}_1 = N \times (c_1 \times c_2 \times \cdots \times c_M) + M + K \tag{8}$$

Table 2 Probability table.

Fault feature	x_1	x_2	$\cdots$	x_M
Referential grades of fault features	$(A_1^1, A_1^2, \cdots, A_1^{c_1})$	$(A_2^1, A_2^2, \cdots, A_2^{c_2})$	$\cdots$	$(A_M^1, A_M^2, \cdots, A_M^{c_M})$
$P_{m,i}^1$	$(P_{1,1}^1, P_{1,2}^1, \cdots, P_{1,c_1}^1)$	$(P_{2,1}^1, P_{2,2}^1, \cdots, P_{2,c_2}^1)$	$\cdots$	$(P_{M,1}^1, P_{M,2}^1, \cdots, P_{M,c_M}^1)$
$P_{m,i}^2$	$(P_{1,1}^2, P_{1,2}^2, \cdots, P_{1,c_1}^2)$	$(P_{2,1}^2, P_{2,2}^2, \cdots, P_{2,c_2}^2)$	$\cdots$	$(P_{M,1}^2, P_{M,2}^2, \cdots, P_{M,c_M}^2)$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$P_{m,i}^N$	$(P_{1,1}^N, P_{1,2}^N, \cdots, P_{1,c_1}^N)$	$(P_{2,1}^N, P_{2,2}^N, \cdots, P_{2,c_2}^N)$	$\cdots$	$(P_{M,1}^N, P_{M,2}^N, \cdots, P_{M,c_M}^N)$

The number of the initialized parameters in BRB-P can be calculated by

$$\mathrm{Num}_2 = N \times (c_1 + c_2 + \cdots + c_M) + M + K \tag{9}$$

A simple example is cited to illustrate the effect of reducing the number of initialized parameters given by experts. It is assumed that there are 6 fault modes and 3 fault features. Each fault feature has 3 referential grades. There are 192 $(6 \times 3^3 + 3 + 3^3 = 192)$ parameters set by experts in the traditional BRB. In comparison, experts only specify 84 $(6 \times (3 + 3 + 3) + 3 + 3^3 = 84)$ parameters in BRB-P. With the number of fault features increasing, the number of the initialized parameters is pictured in Fig. 2.

(2) Establishment of initial BRB-P

The target of transforming a probability table to rules of BRB is to calculate the belief degrees of each rule. This process can be cognitively expressed by Eq. (10), where $\hat{\beta}_{n,k}$ represents the belief degree of the n th fault mode when the input information is the referential values of fault features in the k th rule.

$$\hat{\beta}_{n,k} = P(D_n | (x_1^k \text{ is } A_1^k) \wedge (x_2^k \text{ is } A_2^k) \wedge \cdots \wedge (x_M^k \text{ is } A_M^k)) \tag{10}$$

The ER approach has the advantage of effectively fusing qualitative knowledge and quantitative data. Consequently, factors of the probability table and the weights of fault features are fused into belief degrees of rules in BRB-P by ER. To briefly present the ER fusion process, the first rule is taken as an example to reflect the process from the probability given by experts to belief degrees of rules.

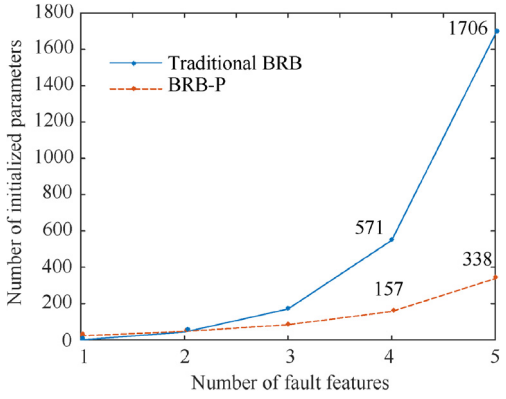


Fig. 2 Comparison of the number of initialized parameters in BRB and BRB-P.

$$R_1: \text{ If } (x_1 \text{ is } A_1^1) \wedge (x_2 \text{ is } A_2^1) \wedge \cdots \wedge (x_M \text{ is } A_M^1),$$
$$\text{then } \{(D_1, \widehat{\beta}_{1,1}), (D_2, \widehat{\beta}_{2,1}), \cdots, (D_N, \widehat{\beta}_{N,1})\}$$
$$\text{with the rule weight } \theta_1 \text{and fault feature weights } \delta_1, \delta_2, \cdots, \delta_M \tag{11}$$

Step 1. Evidence acquisition

The probabilities of $x_i = A_i^1 (i = 1, 2, \cdots, M)$ are picked up as evidence. The component of evidence in the first rule is:

$$P_{m,1}^n = P(x_m = A_m^1 | D_n \text{ happened}) \; m = 1, 2, \cdots, M \; n = 1, 2, \cdots, N \tag{12}$$

which is presented in Table 3. M rows in the table are regarded as M pieces of evidence. For instance, the first row is a piece of evidence corresponding to $x_1 = A_1^1$.

Step 2. Normalization

The probability given by experts is distributed on all the referential grades of a specific fault feature, which satisfies $\sum_{i=1}^{c_m} P_{m,i}^n = 1$ and $\sum_{i=1}^{c_m} P_{m,i}^n = 1$. The symbol "$\not\equiv$" means "not always equal". The probability values should be normalized as the input of ER. The operation of normalization is expressed as

$$P_{m,1}^{i,\prime} = P_{m,1}^i / \sum_{j=1}^{N} P_{m,1}^j \; m = 1, 2, \cdots, M \; i = 1, 2, \cdots, N \tag{13}$$

Step 3. Evidence fusion

The analytical expression of ER algorithm is applied to fuse evidence, shown as follows:

$$\mu = [\sum_{n=1}^{N} \prod_{m=1}^{M} (\delta_m P_{m,1}^{n\prime} + 1 - \delta_m \sum_{i=1}^{N} P_{m,1}^{i\prime}) - (N-1) \prod_{m=1}^{M} (1 - \delta_m \sum_{i=1}^{N} P_{m,1}^{i\prime})]^{-1} \tag{14}$$

$$\hat{\beta}_{j,1} = \frac{\mu \times [\prod_{m=1}^{M} (\delta_m P_{m,1}^{j\prime} + 1 - \delta_m \sum_{i=1}^{N} P_{m,1}^{i\prime}) - \prod_{m=1}^{M} (1 - \delta_m \sum_{i=1}^{N} P_{m,1}^{i\prime})]}{1 - \mu \times [\prod_{m=1}^{M} (1 - \delta_m)]} \tag{15}$$

where μ is an intermediate variable to calculate belief degrees, and $\hat{\beta}_{j,1}$ is the belief degree of D_j in the first rule. According to the above analysis, the belief degrees of the first rule are obtained. Other rules can be generated in the same way. The form of the initial BRB-P is shown in Table 4.

3.2. Reasoning process of BRB-P

In this subsection, the reasoning process of BRB-P is presented based on the initial BRB-P, which includes the transformation of input information, calculation of activation weights, and fusion of activated rules.

Step 1. Transformation of input information

In this paper, the utility-based method is adopted to transform the input information. The quantitative input is trans-

Table 3 Evidence collection.

Fault feature	Referential value of fault feature	Probability value			
		$P^1_{m,1}$	$P^1_{m,1}$		$P^1_{m,1}$
x_1	A^1_1	$P^1_{1,1}$	$P^2_{1,1}$		$P^N_{1,1}$
x_2	A^1_2	$P^1_{2,1}$	$P^2_{2,1}$		$P^N_{2,1}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_M	A^1_M	$P^1_{M,1}$	$P^2_{M,1}$		$P^N_{M,1}$

formed to a belief distribution with belief degrees. The calculation of belief degree for the input I_i of the i th fault feature is expressed as follows:

$$\alpha_{i,j} = (A_i^{j+1} - I_i)/(A_i^{j+1} - A_i^j) \; A_i^j \leqslant I_i \leqslant A_i^{j+1} \tag{16}$$

$$\alpha_{i,j+1} = 1 - \alpha_{i,j}; \; i = 1, 2, \cdots, M; \; j = 1, 2, \cdots, c_i - 1 \tag{17}$$

$$\alpha_{i,s} = 0; \; s = 1, 2, \cdots, c_i \; ; \; s \neq j, j+1 \tag{18}$$

where $\alpha_{i,j}$ and $\alpha_{i,j+1}$ are the belief degrees of I_i corresponding to A_i^j and A_i^{j+1} respectively; A_i^j is the j th referential value of the i th fault feature.

Step 2. Calculation of activation weight

To acquire the activation weights of rules, the normalized combined matching degree α_k in the k th rule should be calculated.[29]

$$\alpha_k = \prod_{i=1}^{M} (\alpha_i^k)^{\bar{\delta}_i} \; \alpha_i^k \in \{\alpha_{i,j}\} \tag{19}$$

$$\bar{\delta}_i = \; \delta_i / \max_{j=1,2,..,M} \{\delta_j\} \text{ so that } 0 \leqslant \bar{\delta}_i \leqslant 1 \; i = 1, 2, \cdots, M \tag{20}$$

The activation weight of each rule is calculated by rule weights and the normalized combined matching degrees. The specific form is expressed as

$$\omega_k = (\theta_k \times \alpha_k) / \sum_{i=1}^{K} \theta_i \alpha_i; \; k = 1, 2, \cdots, K \tag{21}$$

Step 3. Fusion of activated rules

The rule activated by the input is the effective information for diagnostic results. The multiple activated rules are combined by the ER algorithm to generate the final result. The specific process is shown as follows:

$$\mu' = [\sum_{n=1}^{N} \prod_{k=1}^{K} (\omega_k \hat{\beta}_{n,k} + 1 - \omega_k \sum_{i=1}^{N} \hat{\beta}_{i,k}) - (N-1) \prod_{k=1}^{K} (1 - \omega_k \sum_{i=1}^{N} \hat{\beta}_{i,k})]^{-1} \tag{22}$$

Table 4 Rule base of initial BRB-P.

No.	$x_1 \wedge x_2 \wedge \ldots \wedge x_M$	$\{D_1, D_2, \ldots, D_N\}$	θ_k
1	$A_1^1 \wedge A_2^1 \wedge \ldots \wedge A_M^1$	$\{\hat{\beta}_{1,1}, \hat{\beta}_{2,1}, \ldots, \hat{\beta}_{N,1}\}$	θ_1
2	$A_1^2 \wedge A_2^2 \wedge \cdots \wedge A_M^2$	$\{\hat{\beta}_{1,2}, \hat{\beta}_{2,2}, \ldots, \hat{\beta}_{N,2}\}$	θ_2
$\vdots$	$\vdots$	$\vdots$	$\vdots$
K	$A_1^K \wedge A_2^K \wedge \cdots \wedge A_M^K$	$\{\hat{\beta}_{1,K}, \hat{\beta}_{2,K}, \ldots, \hat{\beta}_{N,K}\}$	θ_K

$$\beta_j = \frac{\mu' \times [\prod_{k=1}^{K} (\omega_k \hat{\beta}_{j,k} + 1 - \omega_k \sum_{i=1}^{N} \hat{\beta}_{i,k}) - \prod_{k=1}^{K} (1 - \omega_k \sum_{i=1}^{N} \hat{\beta}_{i,k})]}{1 - \mu' \times [\prod_{k=1}^{K} (1 - \omega_k)]} \tag{23}$$

where β_j represents the belief degree of the j th fault mode. The principle of ER utilized in the process of evidence fusion is essentially the same as that used in the activated rules fusion.

4. Optimization of BRB-P with interpretability constraints

In this section, three interpretability constraints are firstly introduced to maintain the interpretability of BRB-P model in the optimization process.[26,30] Then, the initial BRB-P fault diagnosis model is optimized by P-CMA-ES with three interpretability constraints. The complete process of fault diagnosis based on BRB-P is illustrated at the end of this section.

4.1. Interpretability constraints

The definition and the specific application of three interpretability constraints are illustrated as follows:

(1) The credibility of expert knowledge is introduced to describe the modeling ability of the initial BRB-P.

The information in the probability table is not entirely reliable due to the subjectivity of experts. The optimization process is exploited to improve the accuracy of the fault diagnosis model. The credibility of expert knowledge[26] is utilized to calculate the step size of optimization. The step size of optimization is relatively large when the accuracy of the initial BRB-P model is lower. The search range of the optimization is determined by the step size. The higher the credibility of expert knowledge is, the more accurate the initial BRB-P is, and the smaller the optimization step size is. To represent this relationship, the transformation of Euclidean distance is used to define the credibility of expert knowledge in this paper, which is expressed as

$$\begin{cases} D(\boldsymbol{\beta}, \tilde{\boldsymbol{\beta}}) = \sum_{i=1}^{J} \sqrt{\sum_{j=1}^{N} (\beta_j^i - \tilde{\beta}_j^i)^2} / (N \times J) \\ \boldsymbol{\beta} = [\beta_1^1, \beta_2^1, \cdots, \beta_N^1, \beta_1^2, \cdots, \beta_N^J] \\ \tilde{\boldsymbol{\beta}} = [\tilde{\beta}_1^1, \tilde{\beta}_2^1, \cdots, \tilde{\beta}_N^1, \tilde{\beta}_1^2, \cdots, \tilde{\beta}_N^J] \end{cases} \tag{24}$$

$$K_{\text{ex}} = 1 - D(\boldsymbol{\beta}, \tilde{\boldsymbol{\beta}}), K_{\text{ex}} \in [0, 1] \tag{25}$$

where J is the total number of training samples; β_j^i and $\tilde{\beta}_j^i$ are the diagnostic output of the initial BRB-P model and the practical label respectively; K_{ex} is the credibility of expert knowl-

edge. If $K_{ex}=1$, the initial BRB-P model is completely accurate; if $K_{ex}=0$, the expert knowledge is entirely incredible, and the BRB-P degenerates into a black-box model.

(2) The penalty factor is proposed to enable the optimized BRB-P as close as the initial BRB-P.

The parameters of BRB-P should be fine-tuned in optimization to ensure the effective utilization of the initial judgment of experts.[16] As such, in this paper, the objective function of optimization is improved with the penalty factor. The penalty factor is given by

$$\begin{cases} C(\boldsymbol{\beta}^*,\tilde{\boldsymbol{\beta}}^*)=\sqrt{\sum_{i=1}^{K}\sum_{j=1}^{N}(\beta_{j,i}^*-\tilde{\beta}_{j,i}^*)^2}/K \\ \boldsymbol{\beta}^*=[\beta_{1,1}^*,\beta_{2,1}^*,\cdots,\beta_{N,K}^*] \\ \tilde{\boldsymbol{\beta}}^*=[\tilde{\beta}_{1,1}^*,\tilde{\beta}_{2,1}^*,\cdots,\tilde{\beta}_{N,K}^*] \end{cases} \tag{26}$$

where $\beta_{j,i}^*$ and $\tilde{\beta}_{j,i}^*$ are the belief degrees of initial rules and optimized rules respectively. It follows that the farther the distance between $\boldsymbol{\beta}^*$ and $\tilde{\boldsymbol{\beta}}^*$ is, the greater the penalty factor $C(\boldsymbol{\beta}^*,\tilde{\boldsymbol{\beta}}^*)$ will be.

(3) The rule-activation factor is utilized to mark the activated rules.

The unactivated rules in the training process should preserve the initial judgment given by experts due to the limited fault samples. In this paper, the rule-activation factor W_k is used to express the activation states of the k th rule. If the k th rule has been activated at least once by the training samples, W_k is equal to 1. Otherwise, W_k is equal to 0, and the corresponding rule cannot participate in the optimization process. It should be pointed out that the k th rule is activated when $\omega_k>0.001$.[26]

4.2. Parameter optimization with interpretability constraints

P-CMA-ES algorithm has been well applied to weaken the subjectivity of experts to enhance the accuracy of BRB. However, the purpose of optimization is to improve the accuracy of the model, and the optimized rules are considerably different from the initial rules. Moreover, the parameters may be over-optimized without constraints. Hence, in this paper, the interpretability constraints are introduced to P-CMA-ES to improve the accuracy of the initial BRB-P under the requirement of interpretability. The steps are shown below.

The parameter vector $\boldsymbol{\Psi}$ contains rule weights, fault feature weights and factors of the probability table, shown as follows:

$$\boldsymbol{\Psi}=[\theta_1,\theta_2,\cdots,\theta_K,\delta_1,\delta_2,\cdots,\delta_M,P_{1,1}^1,P_{1,1}^2,\cdots,P_{M,c_M}^N] \tag{27}$$

The optimization objective is given by

$$\begin{aligned} &\max\ f(\boldsymbol{\Psi})-C(\boldsymbol{\beta}^*,\tilde{\boldsymbol{\beta}}^*) \\ &\text{s.t. } 0\leqslant\theta_k\leqslant 1;\ k=1,2,\cdots,K \\ &0\leqslant\delta_j\leqslant 1;\ j=1,2,\cdots,M \\ &\sum_{i=1}^{c_j}P_{j,i}^l=1\ 0\leqslant P_{j,i}^l\leqslant 1;\ l=1,2,\cdots,N \end{aligned} \tag{28}$$

where $f(\boldsymbol{\Psi})$ is the traditional objective function, such as the accuracy rate, mean square error, etc. $C(\boldsymbol{\beta}^*,\tilde{\boldsymbol{\beta}}^*)$ represents the penalty factor. The symbol of objective function depends on the specific application.

Step 1. Initialization operation

The initial parameters of the optimization algorithm are presented in Table 5. The initial values of parameters are given by experts.

Step 2. Sampling operation

$\boldsymbol{\Psi}^0$ is taken as the center point to generate the initial populations with normal distribution:

$$\boldsymbol{\Psi}_i^{g+1}\boldsymbol{w}^g+\varepsilon^g\mathbb{N}(0,\boldsymbol{C}^g)\ i=1,2,\cdots,\lambda \tag{29}$$

where g means the g th generation; $\boldsymbol{\Psi}_i^{g+1}$ is the i th individual of the $(g+1)$ th generation; $\mathbb{N}$ represents the normal distribution.

Step 3. Projection operation

$\boldsymbol{\Psi}_i^{g+1}$ is performed on projection operations one by one to satisfy the equality constraint. The parameters are mapped to the feasible region hyperplane:

$$\begin{aligned} \boldsymbol{\Psi}_i^{g+1}(1+n_e\times(j-1):n_e\times j)=\boldsymbol{\Psi}_i^{g+1}(1+n_e\times(j-1):n_e\times j) \\ -\boldsymbol{A}_e^{\mathrm{T}}\times(\boldsymbol{A}_e\times\boldsymbol{A}_e^{\mathrm{T}})^{-1} \\ \times\boldsymbol{\Psi}_i^{g+1}(1+n_e\times(j-1):n_e\times j)\times\boldsymbol{A}_e \end{aligned} \tag{30}$$

where n_e is the number of parameters of equality constraints; j and $\boldsymbol{A}_e$ represent the number of equality constraints and the equation parameter vector respectively. The hyperplane representing the feasible region of equality constraints is expressed as

$$\boldsymbol{A}_e\boldsymbol{\Psi}_i^{g+1}(1+n_e\times(j-1):n_e\times j)=1 \tag{31}$$

Step 4. Selection operation

The τ optimal populations are selected to update the mean, which makes the direction towards the optimal solution.

$$\boldsymbol{w}^{g+1}=\sum_{i=1}^{\tau}h_i\boldsymbol{\Psi}_{i:\lambda}^{g+1} \tag{32}$$

where h_i is the weight of the i th solution.

Step 5. Updating operation

The covariance matrix $\boldsymbol{C}$ is updated according to the following formula:

$$\begin{aligned} \boldsymbol{C}^{g+1}=(1-c_1-c_2)\boldsymbol{C}^g+c_1\boldsymbol{p}_c^{g+1}(\boldsymbol{p}_c^{g+1})^{\mathrm{T}} \\ +\ c_2\sum_{i=1}^{v}h_i\left(\frac{\boldsymbol{\Psi}_{i:\lambda}^{g+1}-\boldsymbol{w}^g}{\varepsilon^g}\right)\left(\frac{\boldsymbol{\Psi}_{i:\lambda}^{g+1}-\boldsymbol{w}^g}{\varepsilon^g}\right)^{\mathrm{T}} \end{aligned} \tag{33}$$

Table 5 Initial parameters of P-CMA-ES.

Parameter	Meaning	Parameter	Meaning
$\boldsymbol{\Psi}^0$	Initial parameter vector	$\varepsilon^0=1-K_{ex}$	Step size
$\boldsymbol{w}^0=\boldsymbol{\Psi}^0$	Initial mean of population	λ	Population size
$\boldsymbol{C}^0=\boldsymbol{I}$	Initial covariance matrix	τ	Offspring population size
G	Maximum evolution generation	$\boldsymbol{I}$	Unit array

where c_1 and c_2 are the learning rate. $\boldsymbol{p}$ is the evolution path, whose updating rule is

$$\boldsymbol{p}_c^{g+1} = (1-c_c)\boldsymbol{p}_c^{g} + \sqrt{c_c(2-c_c)\left(\sum_{i=1}^{\tau} h_i^2\right)^{-1}} \frac{\boldsymbol{w}^{g+1}-\boldsymbol{w}^{g}}{\varepsilon^{g}} \tag{34}$$

where c_c is the parameter of the evolution path. The step size can be updated as

$$\varepsilon^{g+1} = \varepsilon^{g} \exp\left(\frac{c_\sigma}{d_\sigma}\left(\frac{\| \boldsymbol{p}_\sigma^{g+1} \|}{E\| \mathbb{N}(0,\boldsymbol{I}) \|} - 1\right)\right) \tag{35}$$

$$\boldsymbol{p}_\sigma^{g+1} = (1-c_c)\boldsymbol{p}_\sigma^{g} + \sqrt{c_\sigma(2-c_\sigma)\left(\sum_{i=1}^{\tau} h_i\right)^{-1}} \boldsymbol{C}^{(g)-\frac{1}{2}} \frac{\boldsymbol{w}^{g+1}-\boldsymbol{w}^{g}}{\varepsilon^{g}} \tag{36}$$

where d_σ represents the damping coefficient; $E\| \mathbb{N}(0,\boldsymbol{I}) \|$ and c_σ are the expectation of the Euclidean paradigm $\| \mathbb{N}(0,\boldsymbol{I}) \|$ and the conjugate evolution path parameter respectively; $\boldsymbol{p}_\sigma^{g+1}$ is the conjugate evolution path.

Step 6. Termination criterion

Go back to Step 2 until the maximum evolution generation is achieved.

Step 7. Output optimization result

If $W_k = 1$, the k th rule optimized is preserved. Otherwise, the parameters of the k th rule are reserved.

4.3. Process of fault diagnosis based on BRB-P

In this subsection, the implementation process of the proposed interpretable fault diagnosis method based on BRB-P is presented in Fig. 3. The specific steps are shown as follows:

Step 1. Determination of fault features and fault modes

The fault features and fault modes are the input and output variables in the process of fault diagnosis respectively, which can be selected by the mechanism analysis or the feature extraction method. The referential grades of fault features can be determined by expert experience or specific methods, such as Receiver Operating Characteristic (ROC) curve.

Step 2. Establishment and reasoning of BRB-P fault diagnosis model

The probability table of the system is given by experts based on expert knowledge and historical experience, which has the unified and meaningful form to express the expert knowledge. The factors of the probability table are collected as evidence to generate the belief degrees of rules through the process of normalization and fusion. Then, the input information is transformed into belief distribution of referential grades of fault features by Eqs. (16)–(18). As profiled in Eqs. (19)–(21), the activation weights of rules are calculated. Activated rules are fused to generate the diagnostic results by Eqs. (22)–(23).

Step 3. Optimization of parameters under interpretability requirement

Interpretability constraints are generated based on the practical system. The credibility of expert knowledge is counted based on Eqs. (24)–(25) as the step size of optimization to control the search range; the penalty factor shown as Eq. (26) is employed to construct the new objective function to enable the optimized fault diagnosis model as close as the initial one; the rule-activation factors are calculated based on the training samples to pick out the rules that need to be optimized. The interpretability constraints mentioned above are inserted into the P-CMA-ES optimization algorithm shown as Eqs. (29)–(36). The parameters of the initial BRB-P model are optimized to weaken the expert subjectivity under the interpretability requirement.

Step 4. Output the diagnostic result

So far, the fault diagnosis model based on BRB-P has been established. When the observation data of fault features are input into BRB-P, the diagnostic result is available.

5. Case study

Relay as a control component plays a significant role in signal transmission, load switching, etc. The performance of a relay is considerably essential for the normal operation of the whole system. Taking the relay (JZX-22F) as an example, the fault diagnosis model based on BRB-P is constructed to monitor whether the relay is in a fault state.

5.1. Preparatory work

Some normal relays are divided into six groups. One group is not processed. The other five groups are injected Contact Wear (CW), Adhesion of Normally Open Contact (ANOC), Adhesion of Normally Closed Contact (ANCC), Loose Spring (LS) and Coil Fault (CF). In this subsection, the STS2104A relay test system shown in Fig. 4 is selected to test the above 6 groups of relays.

First, the upper computer, tester, adapter and relay are connected to make up the whole system. Then, the equipment parameters in the upper computer are written according to the electrical parameters of the JZX-22F relay. Finally, the test key is pressed repeatedly to obtain the experimental samples which are stored on the upper computer. 1800 experimental samples have been obtained, in which 600 groups are test samples and the rest are training samples. The training samples are shown in Fig. 5.

In BRB-P model, four variables: Pull-in Voltage (PV), Pull-in Time (PT), Resistance of No Contact (RNC) and Resistance of Contact (RC) are selected as fault features. Three referential grades are assigned for PV, PT, RNC and RC. They are Low (L), Medium (M) and High (H). The referential values of fault features are profiled in Table 6, which are given by experts according to the manual and experimental samples.[26] 6 working states of a relay are shown in Table 7.

5.2. Establishment of initial BRB-P

In this subsection, the initial BRB-P is automatically generated by the probability table given by experts. The first task is to set up the probability table according to the expert knowledge. On the premise of known fault modes and fault features, the probability of the fault feature near a specific referential value when a fault mode occurs is given by experts. The probability table of the relay is shown in Table 8.

The initial BRB-P is established after the process of evidence acquisition, normalization and fusion. The partial rules of the initial BRB-P are shown in Table 9, and the complete rule base is presented in Appendix A.

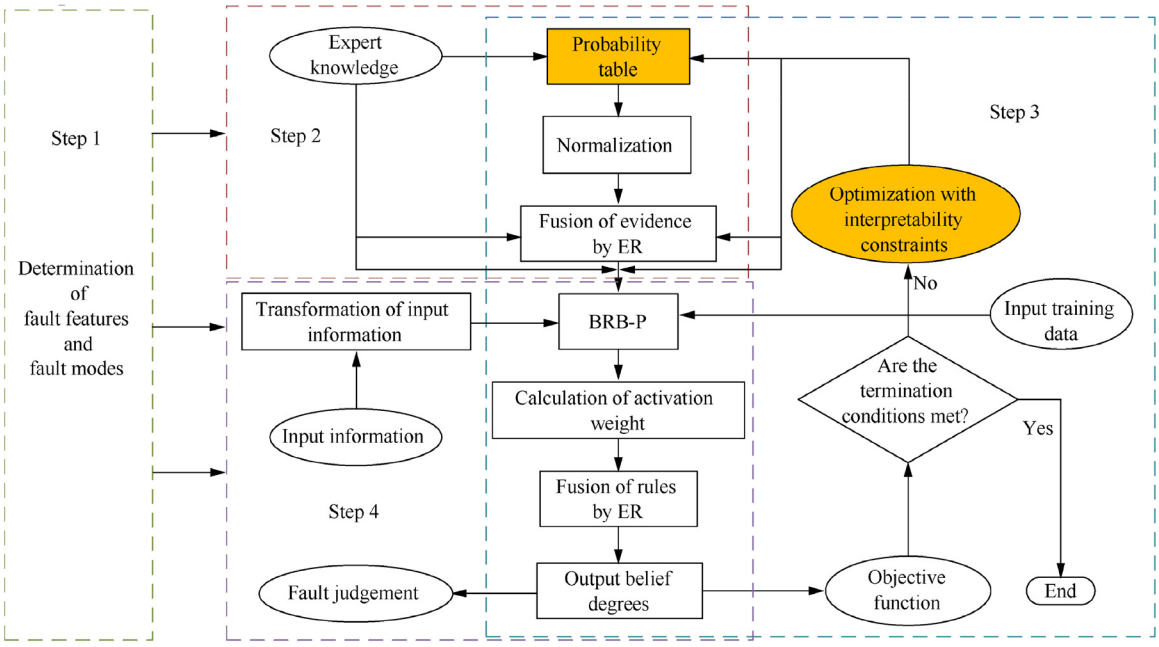


Fig. 3 Process of fault diagnosis based on BRB-P.

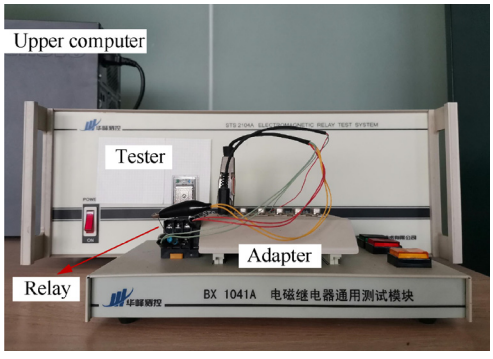


Fig. 4 Relay test system.

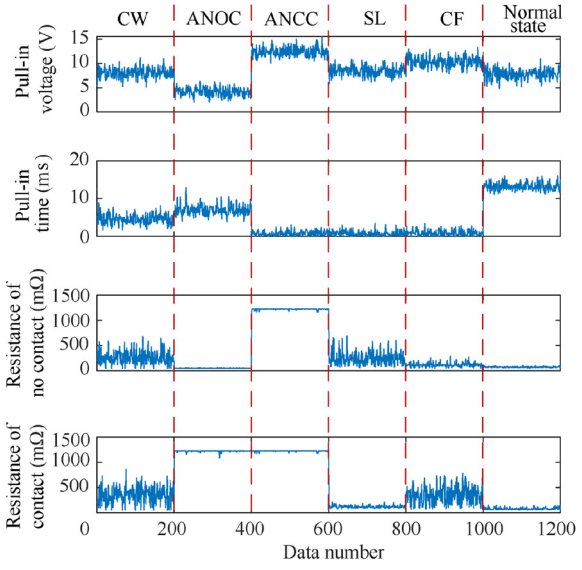


Fig. 5 Training samples of relays.

5.3. Optimization process

To weaken the subjectivity of experts in BRB-P under the premise of ensuring the interpretability of the model, the parameters are optimized by P-CMA-ES with interpretability constraints. The credibility of expert knowledge is calculated as 0.862, which means that the initial search step is 0.138 $(1 - 0.862 = 0.138)$. There are 76 rules activated through the activation judgment. $\sum_{k=1}^{81} W_k = 76$, which means that the remaining 5 rules unactivated maintain the initial judgment of experts. The maximum evolution generation is 200, and the objective function is expressed as Eq. (28). $f(\boldsymbol{\Psi})$ is the accuracy rate of the model. It is also the ratio of the number of correct diagnostic results to the number of total training samples.

The iterative optimization process is shown in Fig. 6.

The partial optimized rules are shown in Table 10, and the complete rule base of BRB-P is given in Appendix B.

Table 6 Referential value of fault feature.

Fault feature	Referential value of fault feature		
	L	*M*	*H*
PV(x_1)(V)	1	7	16
PT(x_2)(ms)	0	5.5	17
RNC(x_3)(mΩ)	10	300	1300
RC(x_4)(mΩ)	10	300	1300

Table 7 Working states of a relay.

Symbol	D_1	D_2	D_3	D_4	D_5	D_6
Working state	CW	ANOC	ANCC	LS	CF	Normal state

Table 8 Probability table of relay fault diagnosis.

Fault feature	PV	PT	RNC	RC
Referential grade	$(L,\ M,\ H)$	$(L,\ M,\ H)$	$(L,\ M,\ H)$	$(L,\ M,\ H)$
$P_{m,i}^1$	(0.1, 0.7, 0.2)	(0.2, 0.8, 0)	(0.3, 0.6, 0.1)	(0.3, 0.6, 0.1)
$P_{m,i}^2$	(0.3, 0.7, 0)	(0, 0.9, 0.1)	(0.85, 0.15, 0)	(0, 0.1, 0.9)
$P_{m,i}^3$	(0, 0.1, 0.9)	(0.95, 0.05, 0)	(0, 0.1, 0.9)	(0, 0.1, 0.9)
$P_{m,i}^4$	(0, 0.7, 0.3)	(0.95, 0.05, 0)	(0.7, 0.3, 0)	(0.7, 0.3, 0)
$P_{m,i}^5$	(0, 0.4, 0.6)	(0.95, 0.05, 0)	(0.65, 0.35, 0)	(0.65, 0.35, 0)
$P_{m,i}^6$	(0.1, 0.7, 0.2)	(0, 0.2, 0.8)	(0.7, 0.3, 0)	(0.7, 0.3, 0)

Table 9 Partial rules of initial BRB-P.

No.	$x_1 \wedge x_2 \wedge x_3 \wedge x_4$	$\{D_1, D_2, D_3, D_4, D_5, D_6\}$	θ_k	No.	$x_1 \wedge x_2 \wedge x_3 \wedge x_4$	$\{D_1, D_2, D_3, D_4, D_5, D_6\}$	θ_k
10	$L \wedge M \wedge L \wedge L$	{0.42,0,0,0,0,0.58}	1	50	$M \wedge H \wedge M \wedge M$	{0,0.02,0,0,0,0.98}	1
20	$L \wedge H \wedge L \wedge M$	{0,0.13,0,0,0,0.87}	1	60	$H \wedge L \wedge M \wedge H$	{0.03,0,0.97,0,0, 0}	1
30	$M \wedge L \wedge L \wedge H$	{1,0,0,0,0,0}	1	70	$H \wedge M \wedge H \wedge L$	{1,0,0,0,0,0}	1
40	$M \wedge M \wedge M \wedge L$	{0.71,0,0,0.05,0.03,0.21}	1	80	$H \wedge H \wedge H \wedge M$	{0,0,0,0,0,1}	1

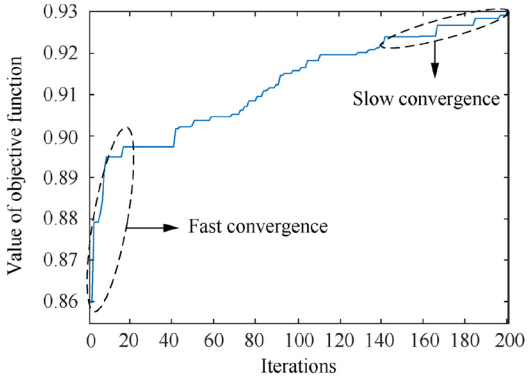


Fig. 6 Iterative process of optimization.

5.4. Comparative study

In this subsection, the traditional BRB, the BRB-P without interpretability constraints, the back propagation neural network and support vector machine are conducted for comparison.

(1) Traditional BRB

According to the expert knowledge and historical experience, the belief degrees, the fault feature weights and the rule weights can be set by experts. In contrast with the proposed method, the initial BRB is established through the traditional method. In other words, 81 rules are provided by experts based on the mechanism in the process of relay fault diagnosis. Please refer to Appendix C for the specific BRB, named BRB0.

(2) BRB-P without interpretability constraints

The initial BRB-P is optimized by the P-CMA-ES without interpretability constraints to verify the effectiveness of interpretability constraints by comparison. The specific parameters of this optimization include: the search step size is 0.5; the maximum evolution generation is 200; the objective function is $f(\boldsymbol{\Psi})$. In this optimization, the parameters of all rules are optimized based on the initial BRB-P. Please refer to Appendix D for the result of optimization, named BRB1.

(3) Back Propagation Neural Network (BPNN) method

Table 10 Partial rules of BRB-P.

No.	$x_1 \wedge x_2 \wedge x_3 \wedge x_4$	$\{D_1, D_2, D_3, D_4, D_5, D_6\}$	θ_k	No.	$x_1 \wedge x_2 \wedge x_3 \wedge x_4$	$\{D_1, D_2, D_3, D_4, D_5, D_6\}$	θ_k
10	$L \wedge M \wedge L \wedge L$	{0.66,0.29,0.01,0.02,0.01,0.01}	0.48	50	$M \wedge H \wedge M \wedge M$	{0,0.01,0,0,0.01,0.98}	0.66
20	$L \wedge H \wedge L \wedge M$	{0.04,0.15,0,0,0.04,0.77}	0.31	60	$H \wedge L \wedge M \wedge H$	{0.05,0.04,0.52,0.17,0.2,0.02}	0.56
30	$M \wedge L \wedge L \wedge H$	{0.06,0.42,0.22,0.18,0.12,0}	0.54	70	$H \wedge M \wedge H \wedge L$	{0.54,0.02,0.19,0.1,0.09,0.06}	0.94
40	$M \wedge M \wedge M \wedge L$	{0.38,0.11,0.01,0.45,0.01,0.04}	0.69	80	$H \wedge H \wedge H \wedge M$	{0,0,0,0,0.03,0.97}	0.86

Table 11 Parameters of BPNN.

Parameter	Time of training(time)	Training goal	Learning rate
Value	10000	0.0001	0.04

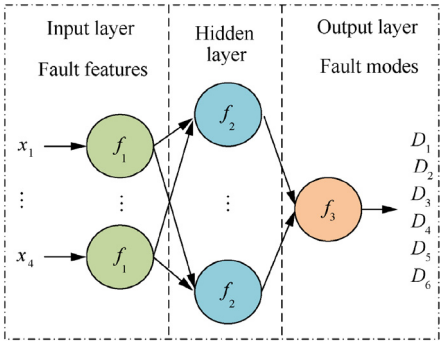


Fig. 7 Structure of BPNN.

Table 12 Accuracy comparison of different models.

Part	Model	Accuracy rate(%)
Part A	BRB0	73.67
	Initial BRB-P	**87.33**
Part B	BRB1	97.5
	BRB-P	**97.33**
	BPNN	93.67
	SVM	97

BPNN is used to establish the relationship of fault features and fault modes, which is a typical black-box algorithm. The structure of BPNN is shown in Fig. 7, where $f_i(i = 1, 2, 3)$ is the transfer function. The structure of BPNN includes the input layer, hidden layer and output layer, which are connected through weight parameters. Similarly, the values of fault features are regarded as the input of the neural network model. However, the output is one of the working states of the relay represented by a digital label. In the training process, the weight parameters are constantly updated to obtain better diagnostic results. The parameters of BPNN are shown in Table 11.

(4) Support Vector Machine (SVM) method

SVM is a typical method used in discriminative model, which has the advantages of less dependence on samples and preventing overfitting. The main target of SVM is to search the classification hyperplane to classify samples. This experiment is carried out with the help of the LibSVM toolbox. Radial Basis Function (RBF) is adopted as kernel function.

5.5. Result analysis

In this subsection, the comparative study of the initial BRB-P, BRB0, BRB-P, BRB1, BPNN and SVM is made to analyze the accuracy and interpretability of models.

(1) Analysis of model accuracy and the number of initialized parameters

The accuracy of the models mentioned above is concluded in Table 12, which is divided into two parts.

In Part A, the accuracy of the initial BRB-P is desirable. Less subjectivity of experts is contained in the initial BRB-P owing to fewer parameters given by experts. The accuracy of the initial BRB-P is higher than that of BRB0. The results of the relay fault diagnosis in Part A are shown in Fig. 8. By comparing BRB0 with BRB-P, the number of initialized parameters reduces from 571 to 157, calculated by Eqs. (8)–(9). It can be seen that the number of parameters in BRB-P is reduced, and the accuracy of BRB-P is effectively enhanced by introducing the probability table.

The diagnostic results of Part B are shown in Fig. 9. Although BPNN and SVM can achieve high accuracy, they are data-driven models whose reasoning processes cannot be understood intuitively. It is obvious that the modeling ability of SVM is better than that of BPNN when there are limited fault samples. Initializing the parameters by experts releases the burden of optimization, and the accuracy of the optimized models based on BRB-P is higher in these models. Compared with BRB1, the accuracy of BRB-P is decreased due to the constrained optimization process. However, more importantly,

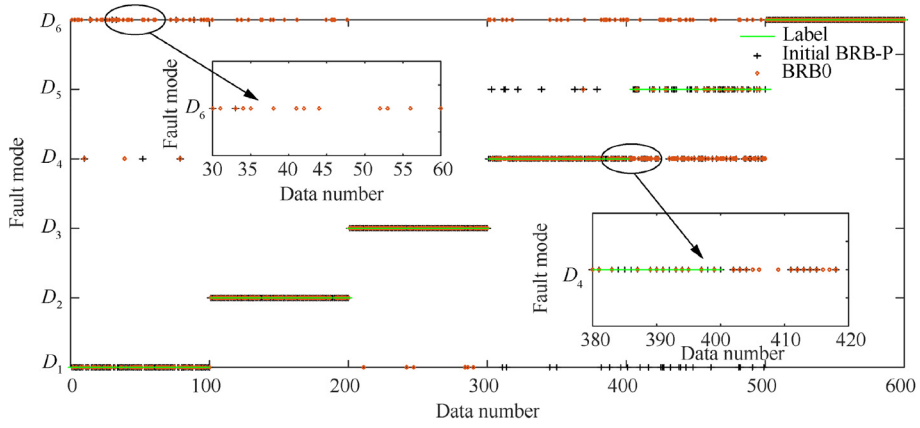


Fig. 8 Comparison of model accuracy in Part A.

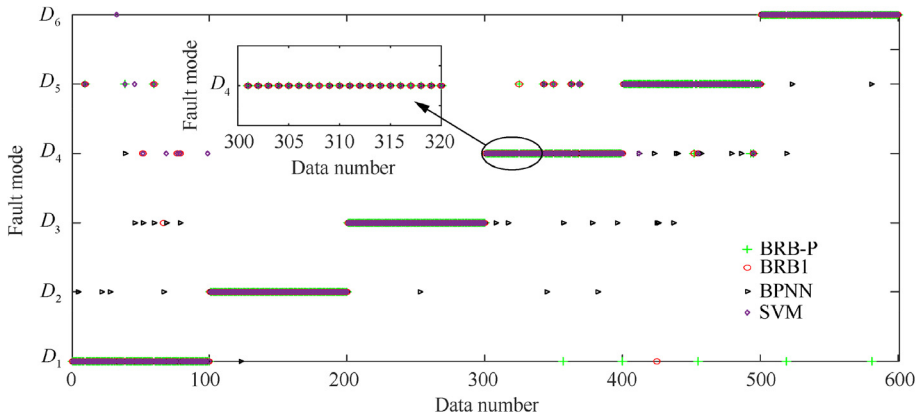


Fig. 9 Comparison of model accuracy in Part B.

Table 13 Euclidean distance of belief degrees for different models.

Model	Euclidean distance
BRB1 and initial BRB-P	6.80
BRB-P and initial BRB-P	4.56

the interpretability of the model is retained under the premise of the same maximum evolution generation. The specific analysis of interpretability is presented below.

(2) Analysis of model interpretability

BRB is essentially an interpretable model. However, the excellent property may be destroyed in the optimization process, which mainly reflects in two aspects: the deviation from the initial judgment of experts and over-optimization of parameters. The following will focus on these two aspects.

$$d(\boldsymbol{\beta}^*, \tilde{\boldsymbol{\beta}}^*) = \sqrt{\sum_{i=1}^{81}\sum_{j=1}^{6}(\beta_{j,i}^* - \tilde{\beta}_{j,i}^*)^2} \tag{37}$$

To avoid deviating from experts' initial judgment, the optimized parameters are required to be close to the initial BRB-P. As can be seen from Tables 9 and 10, the parameters of rules are fine-tuned based on the initial rules. In other words, although the parameters are optimized to improve the accuracy, most of the rules are still in line with the cognition of experts. More specifically, the Euclidean distance shown as Eq. (37) is introduced to describe this closeness. From Table 13, compared with BRB1, the Euclidean distance between the initial BRB-P and BRB-P is shorter, which indicates that the result of optimization with interpretability constraints is closer to the initial judgment of experts.

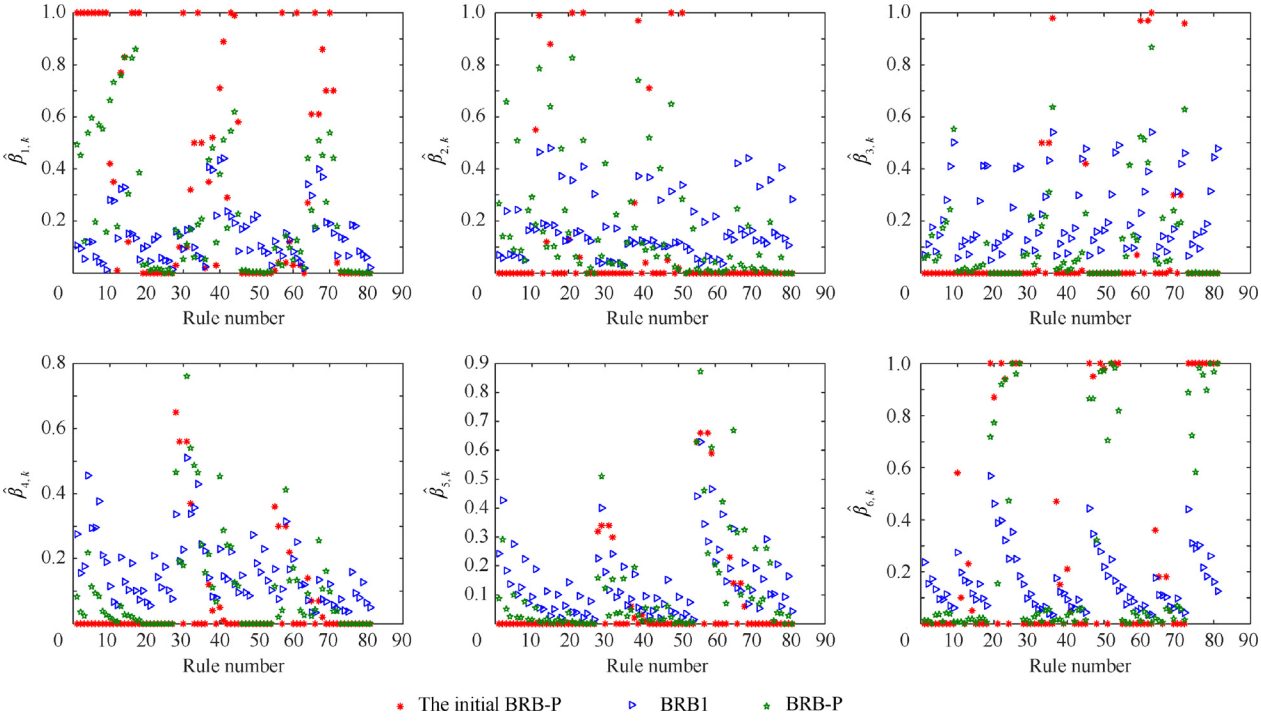


Fig. 10 Belief degrees of rules.

To further reveal the level of closeness, the belief degrees of models and the difference between them are drawn in Figs. 10 and 11 respectively. It can be seen from Fig. 10 that the belief degrees of BRB-P are closer to the belief degrees of the initial BRB-P than those of the BRB1. Fig. 11 also presents that the differences between the initial belief degrees and the optimized belief degrees of BRB-P are closer to 0.

To avoid the over-optimization of parameters, the rule-activation factor is proposed. The value of the rule-activation factor is shown in Fig. 12. The sum of W_k is 76, which means that the remaining 5 rules are not involved in the fusion process.

The unactivated rules are usually rare conditions or bad inputs. To ensure the completeness of the rule base, the rules in all cases are set by experts to deal with new situations or error signals. Here, 5 unactivated rules are presented in Table 14, in which working states of relays are recognized as normal states by experts. The rule-activation factor is introduced into the optimization process to judge whether a rule is activated, so as to determine whether to optimize the rule.

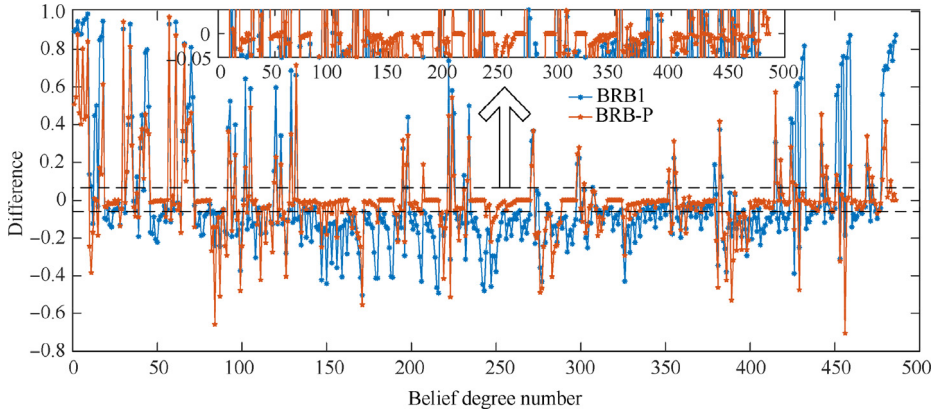


Fig. 11 Difference of belief degrees.

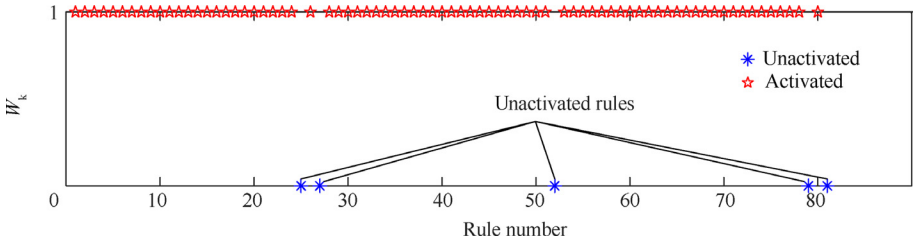


Fig. 12 Rule-activation factor.

Table 14 Unactivated rules.

No.	$x_1 \wedge x_2 \wedge x_3 \wedge x_4$	Model	$\{D_1, D_2, D_3, D_4, D_5, D_6\}$	θ_k
25	$L \wedge H \wedge H \wedge L$	Initial BRB-P	{0,0,0,0,0,1}	1
		BRB-P	{0,0,0,0,0,1}	1
		BRB1	{0.05,0.14,0.25,0.17,0.04,0.35}	0.57
27	$L \wedge H \wedge H \wedge H$	Initial BRB-P	{0,0,0,0,0,1}	1
		BRB-P	{0,0,0,0,0,1}	1
		BRB1	{0.01,0.3,0.41,0.08,0.01,0.19}	0.45
52	$M \wedge H \wedge H \wedge L$	Initial BRB-P	{0,0,0,0,0,1}	1
		BRB-P	{0,0,0,0,0,1}	1
		BRB1	{0.08,0.1,0.3,0.23,0.03,0.26}	0.78
79	$H \wedge H \wedge H \wedge L$	Initial BRB-P	{0,0,0,0,0,1}	1
		BRB-P	{0,0,0,0,0,1}	1
		BRB1	{0.07,0.13,0.31,0.13,0.09,0.27}	0.77
81	$H \wedge H \wedge H \wedge H$	Initial BRB-P	{0,0,0,0,0,1}	1
		BRB-P	{0,0,0,0,0,1}	1
		BRB1	{0.02,0.28,0.48,0.05,0.04,0.13}	0.84

Table A1 Weights of fault features in initial BRB-P.

Fault feature	PV	PT	RNC	RC
Fault feature weight	1	1	1	1

In the fault diagnosis model based on BRB-P, the parameters of the 5 rules mentioned above are not optimized and reserve the initial judgment given by experts to avoid the over-optimization of parameters. However, these rules are updated

Table A2 Rule base of initial BRB-P.

No.	$x_1 \wedge x_2 \wedge x_3 \wedge x_4$	$\{D_1, D_2, D_3, D_4, D_5, D_6\}$	θ_k	No.	$x_1 \wedge x_2 \wedge x_3 \wedge x_4$	$\{D_1, D_2, D_3, D_4, D_5, D_6\}$	θ_k
1	$L \wedge L \wedge L \wedge L$	{1,0,0,0,0,0}	1	42	$M \wedge M \wedge M \wedge H$	{0.28,0.72,0,0,0,0}	1
2	$L \wedge L \wedge L \wedge M$	{1,0,0,0,0,0}	1	43	$M \wedge M \wedge H \wedge L$	{1,0,0,0,0,0}	1
3	$L \wedge L \wedge L \wedge H$	{1,0,0,0,0,0}	1	44	$M \wedge M \wedge H \wedge M$	{0.99,0,0.01,0,0,0}	1
4	$L \wedge L \wedge M \wedge L$	{1,0,0,0,0,0}	1	45	$M \wedge M \wedge H \wedge H$	{0.58,0,0.42,0,0,0}	1
5	$L \wedge L \wedge M \wedge M$	{1,0,0,0,0,0}	1	46	$M \wedge H \wedge L \wedge L$	{0,0,0,0,0,1}	1
6	$L \wedge L \wedge M \wedge H$	{1,0,0,0,0,0}	1	47	$M \wedge H \wedge L \wedge M$	{0,0.05,0,0,0,0.95}	1
7	$L \wedge L \wedge H \wedge L$	{1,0,0,0,0,0}	1	48	$M \wedge H \wedge L \wedge H$	{0,1,0,0,0,0}	1
8	$L \wedge L \wedge H \wedge M$	{1,0,0,0,0,0}	1	49	$M \wedge H \wedge M \wedge L$	{0,0,0,0,0,1}	1
9	$L \wedge L \wedge H \wedge H$	{1,0,0,0,0,0}	1	50	$M \wedge H \wedge M \wedge M$	{0,0.02,0,0,0,0.98}	1
10	$L \wedge M \wedge L \wedge L$	{0.42,0,0,0,0,0.58}	1	51	$M \wedge H \wedge M \wedge H$	{0,1,0,0,0,0}	1
11	$L \wedge M \wedge L \wedge M$	{0.35,0.55,0,0,0,0.1}	1	52	$M \wedge H \wedge H \wedge L$	{0,0,0,0,0,1}	1
12	$L \wedge M \wedge L \wedge H$	{0.01,0.99,0,0,0,0}	1	53	$M \wedge H \wedge H \wedge M$	{0,0,0,0,0,1}	1
13	$L \wedge M \wedge M \wedge L$	{0.77,0,0,0,0,0.23}	1	54	$M \wedge H \wedge H \wedge H$	{0,0,0,0,0,1}	1
14	$L \wedge M \wedge M \wedge M$	{0.83,0.12,0,0,0,0.05}	1	55	$H \wedge L \wedge L \wedge L$	{0.01,0,0,0.36,0.63,0}	1
15	$L \wedge M \wedge M \wedge H$	{0.12,0.88,0,0,0,0}	1	56	$H \wedge L \wedge L \wedge M$	{0.04,0,0,0.3,0.66,0}	1
16	$L \wedge M \wedge H \wedge L$	{1,0,0,0,0,0}	1	57	$H \wedge L \wedge L \wedge H$	{1,0,0,0,0,0}	1
17	$L \wedge M \wedge H \wedge M$	{1,0,0,0,0,0}	1	58	$H \wedge L \wedge M \wedge L$	{0.04,0,0,0.3,0.66,0}	1
18	$L \wedge M \wedge H \wedge H$	{1,0,0,0,0,0}	1	59	$H \wedge L \wedge M \wedge M$	{0.12,0,0.07,0.22,0.59,0}	1
19	$L \wedge H \wedge L \wedge L$	{0,0,0,0,0,1}	1	60	$H \wedge L \wedge M \wedge H$	{0.03,0,0.97,0,0,0}	1
20	$L \wedge H \wedge L \wedge M$	{0,0.13,0,0,0,0.87}	1	61	$H \wedge L \wedge H \wedge L$	{1,0,0,0,0,0}	1
21	$L \wedge H \wedge L \wedge H$	{0,1,0,0,0,0}	1	62	$H \wedge L \wedge H \wedge M$	{0.03,0,0.97,0,0,0}	1
22	$L \wedge H \wedge M \wedge L$	{0,0,0,0,0,1}	1	63	$H \wedge L \wedge H \wedge H$	{0,0,1,0,0,0}	1
23	$L \wedge H \wedge M \wedge M$	{0,0.06,0,0,0,0.94}	1	64	$H \wedge M \wedge L \wedge L$	{0.27,0,0,0.14,0.23,0.36}	1
24	$L \wedge H \wedge M \wedge H$	{0,1,0,0,0,0}	1	65	$H \wedge M \wedge L \wedge M$	{0.61,0,0,0.07,0.14,0.18}	1
25	$L \wedge H \wedge H \wedge L$	{0,0,0,0,0,1}	1	66	$H \wedge M \wedge L \wedge H$	{1,0,0,0,0,0}	1
26	$L \wedge H \wedge H \wedge M$	{0,0,0,0,0,1}	1	67	$H \wedge M \wedge M \wedge L$	{0.61,0,0,0.07,0.14,0.18}	1
27	$L \wedge H \wedge H \wedge H$	{0,0,0,0,0,1}	1	68	$H \wedge M \wedge M \wedge M$	{0.86,0,0.01,0.02,0.06,0.05}	1
28	$M \wedge L \wedge L \wedge L$	{0.03,0,0,0.65,0.32,0}	1	69	$H \wedge M \wedge M \wedge H$	{0.7,0,0.3,0,0,0}	1
29	$M \wedge L \wedge L \wedge M$	{0.1,0,0,0.56,0.34,0}	1	70	$H \wedge M \wedge H \wedge L$	{1,0,0,0,0,0}	1
30	$M \wedge L \wedge L \wedge H$	{1,0,0,0,0,0}	1	71	$H \wedge M \wedge H \wedge M$	{0.7,0,0.3,0,0,0}	1
31	$M \wedge L \wedge M \wedge L$	{0.1,0,0,0.56,0.34,0}	1	72	$H \wedge M \wedge H \wedge H$	{0.04,0,0.96,0,0,0}	1
32	$M \wedge L \wedge M \wedge M$	{0.32,0,0.01,0.37,0.3,0}	1	73	$H \wedge H \wedge L \wedge L$	{0,0,0,0,0,1}	1
33	$M \wedge L \wedge M \wedge H$	{0.5,0,0.5,0,0,0}	1	74	$H \wedge H \wedge L \wedge M$	{0,0,0,0,0,1}	1
34	$M \wedge L \wedge H \wedge L$	{1,0,0,0,0,0}	1	75	$H \wedge H \wedge L \wedge H$	{0,0,0,0,0,1}	1
35	$M \wedge L \wedge H \wedge M$	{0.5,0,0.5,0,0,0}	1	76	$H \wedge H \wedge M \wedge L$	{0,0,0,0,0,1}	1
36	$M \wedge L \wedge H \wedge H$	{0.02,0,0.98,0,0,0}	1	77	$H \wedge H \wedge M \wedge M$	{0,0,0,0,0,1}	1
37	$M \wedge M \wedge L \wedge L$	{0.35,0,0,0.12,0.06,0.47}	1	78	$H \wedge H \wedge M \wedge H$	{0,0,0,0,0,1}	1
38	$M \wedge M \wedge L \wedge M$	{0.52,0.27,0,0.04,0.02,0.15}	1	79	$H \wedge H \wedge H \wedge L$	{0,0,0,0,0,1}	1
39	$M \wedge M \wedge L \wedge H$	{0.03,0.97,0,0,0,0}	1	80	$H \wedge H \wedge H \wedge M$	{0,0,0,0,0,1}	1
40	$M \wedge M \wedge M \wedge L$	{0.71,0,0,0.05,0.03,0.21}	1	81	$H \wedge H \wedge H \wedge H$	{0,0,0,0,0,1}	1
41	$M \wedge M \wedge M \wedge M$	{0.89,0.04,0,0.01,0.01,0.05}	1				

Table B1 Probability table of BRB-P.

Fault feature	PV	PT	RNC	RC
Referential grade	(L, M, H)	(L, M, H)	(L, M, H)	(L, M, H)
D_1	(0.4, 0.39, 0.21)	(0.4, 0.6, 0)	(0.28, 0.5, 0.22)	(0.41, 0.39, 0.2)
D_2	(0.39, 0.56, 0.05)	(0.4, 0.49, 0.11)	(0.66, 0.33, 0.01)	(0.08, 0.01, 0.91)
D_3	(0.09, 0.3, 0.61)	(0.89, 0.11, 0)	(0.15, 0.13, 0.72)	(0.07, 0.1, 0.83)
D_4	(0.04, 0.79, 0.17)	(0.79, 0.21, 0)	(0.06, 0.9, 0.04)	(0.49, 0.19, 0.32)
D_5	(0.03, 0.18, 0.79)	(0.83, 0.14, 0.03)	(0.89, 0.11, 0)	(0.13, 0.66, 0.21)
D_6	(0.13, 0.5, 0.37)	(0.06, 0.03, 0.91)	(0, 0.65, 0.35)	(0.46, 0.43, 0.11)

in BRB1 without basis. Obviously, the optimized results for these 5 rules are irrational in BRB1. For example, the diagnostic results of BRB-P and the initial BRB-P are that the relay is in normal state when the input is consistent with the antecedent of the 27th rule. However, the diagnostic result of BRB1 is the adhesion of normally closed contact in the same case. Consequently, the over-optimization of parameters is avoided, and the interpretability of fault diagnosis model is preserved in this proposed method.

Table B2 Weights of fault feature in BRB-P.

Fault feature	PV	PT	RNC	RC
Fault feature weight	0.99	1	0.88	0.96

6. Conclusions

Fault diagnosis plays an irreplaceable role in the operation and maintenance of the system. An interpretable fault diagnosis approach can build good trust between the model and human beings. As an interpretable expert system, BRB has been widely applied in the field of fault diagnosis.

In this paper, a new interpretable fault diagnosis model based on BRB-P is proposed, where the probability table is exploited to transform the expert knowledge into rules of BRB-P. Three interpretability constraints are inserted into the P-CMA-ES optimization algorithm to improve the accuracy of the model under the requirement of interpretability. Moreover, the fault diagnosis model of the JZX-22F relay based on BRB-P is established. The accuracy and interpretability of BRB-P model can be maintained by comparing with other fault diagnosis models.

Table B3 Rule base of BRB-P.

No.	$x_1 \wedge x_2 \wedge x_3 \wedge x_4$	$\{D_1, D_2, D_3, D_4, D_5, D_6\}$	θ_k	No.	$x_1 \wedge x_2 \wedge x_3 \wedge x_4$	$\{D_1, D_2, D_3, D_4, D_5, D_6\}$	θ_k
1	$L \wedge L \wedge L \wedge L$	{0.49,0.27,0.06,0.08,0.09,0.01}	0.39	42	$M \wedge M \wedge M \wedge H$	{0.17,0.52,0.04,0.24,0.01,0.02}	0.56
2	$L \wedge L \wedge L \wedge M$	{0.45,0.14,0.07,0.04,0.29,0.01}	0.66	43	$M \wedge M \wedge H \wedge L$	{0.55,0.1,0.05,0.24,0.01,0.05}	0.72
3	$L \wedge L \wedge L \wedge H$	{0.12,0.66,0.14,0.03,0.05,0}	0.68	44	$M \wedge M \wedge H \wedge M$	{0.62,0.06,0.08,0.13,0.05,0.06}	0.77
4	$L \wedge L \wedge M \wedge L$	{0.54,0.14,0.05,0.22,0.02,0.03}	0.93	45	$M \wedge M \wedge H \wedge H$	{0.23,0.4,0.23,0.12,0.01,0.01}	0.4
5	$L \wedge L \wedge M \wedge M$	{0.6,0.09,0.07,0.1,0.1,0.04}	0.6	46	$M \wedge H \wedge L \wedge L$	{0,0.11,0,0,0.02,0.87}	0.59
6	$L \wedge L \wedge M \wedge H$	{0.2,0.51,0.17,0.09,0.02,0.01}	0.66	47	$M \wedge H \wedge L \wedge M$	{0.01,0.06,0,0,0.06,0.87}	0.48
7	$L \wedge L \wedge H \wedge L$	{0.57,0.09,0.2,0.08,0.02,0.04}	0.17	48	$M \wedge H \wedge L \wedge H$	{0.01,0.65,0,0,0.02,0.32}	0.7
8	$L \wedge L \wedge H \wedge M$	{0.55,0.05,0.24,0.04,0.08,0.04}	0.4	49	$M \wedge H \wedge M \wedge L$	{0,0.02,0,0,0,0.98}	0.92
9	$L \wedge L \wedge H \wedge H$	{0.16,0.24,0.55,0.03,0.01,0.01}	0.72	50	$M \wedge H \wedge M \wedge M$	{0,0.01,0,0,0.01,0.98}	0.66
10	$L \wedge M \wedge L \wedge L$	{0.66,0.29,0.01,0.02,0.01,0.01}	0.48	51	$M \wedge H \wedge M \wedge H$	{0.01,0.28,0,0,0.01,0.7}	0.46
11	$L \wedge M \wedge L \wedge M$	{0.73,0.19,0.01,0.01,0.05,0.01}	0.59	52	$M \wedge H \wedge H \wedge L$	{0,0,0,0,0,1}	1
12	$L \wedge M \wedge L \wedge H$	{0.18,0.79,0.02,0.01,0,0}	0.98	53	$M \wedge H \wedge H \wedge M$	{0,0.01,0,0,0.01,0.98}	0.55
13	$L \wedge M \wedge M \wedge L$	{0.76,0.16,0.01,0.05,0,0.02}	0.28	54	$M \wedge H \wedge H \wedge H$	{0.01,0.17,0,0,0,0.82}	0.04
14	$L \wedge M \wedge M \wedge M$	{0.83,0.1,0,0.03,0.02,0.02}	0.98	55	$H \wedge L \wedge L \wedge L$	{0.1,0.02,0.13,0.11,0.63,0.01}	0.43
15	$L \wedge M \wedge M \wedge H$	{0.3,0.64,0.02,0.04,0,0}	0.93	56	$H \wedge L \wedge L \wedge M$	{0.04,0,0.06,0.02,0.88,0}	0.85
16	$L \wedge M \wedge H \wedge L$	{0.83,0.1,0.02,0.02,0,0.03}	0.88	57	$H \wedge L \wedge L \wedge H$	{0.03,0.05,0.41,0.04,0.47,0}	0.66
17	$L \wedge M \wedge H \wedge M$	{0.86,0.06,0.03,0.01,0.02,0.02}	0.81	58	$H \wedge L \wedge M \wedge L$	{0.14,0.01,0.15,0.41,0.24,0.05}	0.88
18	$L \wedge M \wedge H \wedge H$	{0.39,0.48,0.12,0,0,0.01}	0.86	59	$H \wedge L \wedge M \wedge M$	{0.1,0,0.13,0.13,0.61,0.03}	0.86
19	$L \wedge H \wedge L \wedge L$	{0.03,0.24,0,0,0.01,0.72}	0.33	60	$H \wedge L \wedge M \wedge H$	{0.05,0.04,0.52,0.17,0.2,0.02}	0.56
20	$L \wedge H \wedge L \wedge M$	{0.04,0.15,0,0,0.04,0.77}	0.31	61	$H \wedge L \wedge H \wedge L$	{0.13,0.01,0.51,0.13,0.18,0.04}	0.94
21	$L \wedge H \wedge L \wedge H$	{0.01,0.83,0,0,0.01,0.15}	0.48	62	$H \wedge L \wedge H \wedge M$	{0.08,0,0.42,0.04,0.42,0.04}	0.56
22	$L \wedge H \wedge M \wedge L$	{0.02,0.06,0,0,0,0.92}	0.47	63	$H \wedge L \wedge H \wedge H$	{0.02,0.01,0.87,0.03,0.07,0}	0.57
23	$L \wedge H \wedge M \wedge M$	{0.02,0.04,0,0,0,0.94}	0.29	64	$H \wedge M \wedge L \wedge L$	{0.44,0.06,0.05,0.09,0.33,0.03}	0.85
24	$L \wedge H \wedge M \wedge H$	{0.02,0.51,0,0,0,0.47}	0.98	65	$H \wedge M \wedge L \wedge M$	{0.24,0.02,0.04,0.02,0.67,0.01}	0.94
25	$L \wedge H \wedge H \wedge L$	{0,0,0,0,0,1}	1	66	$H \wedge M \wedge L \wedge H$	{0.18,0.25,0.21,0.04,0.32,0}	0.51
26	$L \wedge H \wedge H \wedge M$	{0.02,0.02,0,0,0,0.96}	0.8	67	$H \wedge M \wedge M \wedge L$	{0.51,0.03,0.04,0.26,0.1,0.06}	0.4
27	$L \wedge H \wedge H \wedge H$	{0,0,0,0,0,1}	1	68	$H \wedge M \wedge M \wedge M$	{0.45,0.02,0.05,0.11,0.32,0.05}	0.95
28	$M \wedge L \wedge L \wedge L$	{0.17,0.13,0.07,0.47,0.16,0}	0.22	69	$H \wedge M \wedge M \wedge H$	{0.27,0.18,0.24,0.16,0.13,0.02}	1
29	$M \wedge L \wedge L \wedge M$	{0.14,0.08,0.07,0.19,0.51,0.01}	0.59	70	$H \wedge M \wedge H \wedge L$	{0.54,0.02,0.19,0.1,0.09,0.06}	0.94
30	$M \wedge L \wedge L \wedge H$	{0.06,0.42,0.22,0.18,0.12,0}	0.54	71	$H \wedge M \wedge H \wedge M$	{0.44,0.01,0.19,0.04,0.26,0.06}	0.8
31	$M \wedge L \wedge M \wedge L$	{0.11,0.04,0.03,0.76,0.03,0.03}	0.78	72	$H \wedge M \wedge H \wedge H$	{0.18,0.07,0.63,0.04,0.07,0.01}	0.89
32	$M \wedge L \wedge M \wedge M$	{0.17,0.04,0.06,0.54,0.15,0.04}	0.15	73	$H \wedge H \wedge L \wedge L$	{0.01,0.02,0,0,0.08,0.89}	0.35
33	$M \wedge L \wedge M \wedge H$	{0.06,0.23,0.18,0.49,0.04,0}	0.17	74	$H \wedge H \wedge L \wedge M$	{0.01,0.01,0,0,0.26,0.72}	0.71
34	$M \wedge L \wedge H \wedge L$	{0.19,0.04,0.22,0.47,0.04,0.04}	0.68	75	$H \wedge H \wedge L \wedge H$	{0.01,0.2,0,0,0.21,0.58}	0.71
35	$M \wedge L \wedge H \wedge M$	{0.21,0.02,0.31,0.24,0.16,0.06}	0.69	76	$H \wedge H \wedge M \wedge L$	{0,0,0,0,0.01,0.99}	0.68
36	$M \wedge L \wedge H \wedge H$	{0.05,0.12,0.64,0.16,0.03,0}	0.8	77	$H \wedge H \wedge M \wedge M$	{0,0,0,0,0.04,0.96}	0.52
37	$M \wedge M \wedge L \wedge L$	{0.43,0.27,0.01,0.21,0.05,0.03}	1	78	$H \wedge H \wedge M \wedge H$	{0,0.06,0,0,0.04,0.9}	0.89
38	$M \wedge M \wedge L \wedge M$	{0.48,0.18,0.02,0.11,0.19,0.02}	0.77	79	$H \wedge H \wedge H \wedge L$	{0,0,0,0,0,1}	1
39	$M \wedge M \wedge L \wedge H$	{0.12,0.74,0.04,0.07,0.03,0}	0.3	80	$H \wedge H \wedge H \wedge M$	{0,0,0,0,0.03,0.97}	0.86
40	$M \wedge M \wedge M \wedge L$	{0.38,0.11,0.01,0.45,0.01,0.04}	0.69	81	$H \wedge H \wedge H \wedge H$	{0,0,0,0,0,1}	1
41	$M \wedge M \wedge M \wedge M$	{0.51,0.09,0.02,0.29,0.05,0.04}	0.8				

Table C1 Weights of fault feature in BRB0.

Fault feature	PV	PT	RNC	RC
Fault feature weight	1	1	1	1

Table D1 Weights of fault features in BRB1.

Fault feature	PV	PT	RNC	RC
Fault feature weight	0.78	0.82	0.85	0.8

There are mainly two contributions in this paper. On the one hand, the probability table is proposed to express the expert knowledge and establish the initial BRB-P, which reduces the number of initialized parameters under the premise of higher accuracy. The problem of embedding expert knowledge to BRB when many fault features have multiple referential grades is solved. On the other hand, three interpretability constraints are introduced to P-CMA-ES. The credibility of expert knowledge can describe the modeling ability of the initial BRB-P. The penalty factor can enable the optimized BRB-P as close as the initial BRB-P. The problems of the deviation from the experts' initial judgment can be solved by these two interpretability constraints. The rule-activation factor is utilized to mark the activated rules during the training process, which avoids over-optimization of parameters. The interpretability of BRB-P is maintained in optimization.

In this paper, a new interpretable fault diagnosis model based on expert knowledge is proposed. Many data-based models are used in the industry when mass fault samples are available. How to establish an interpretable fault diagnosis model in terms of mass fault samples and how to conduct the performance analysis of BRB-P should be further discussed. The real-time data can be disturbed by the actual environment and the reliability of sensors in practical applications.

Table C2 Rule base of BRB0.

No.	$x_1 \wedge x_2 \wedge x_3 \wedge x_4$	$\{D_1, D_2, D_3, D_4, D_5, D_6\}$	θ_k	No.	$x_1 \wedge x_2 \wedge x_3 \wedge x_4$	$\{D_1, D_2, D_3, D_4, D_5, D_6\}$	θ_k
1	$L \wedge L \wedge L \wedge L$	{1,0,0,0,0,0}	1	42	$M \wedge M \wedge M \wedge H$	{0.6,0,0,0,0,0.4}	1
2	$L \wedge L \wedge L \wedge M$	{1,0,0,0,0,0}	1	43	$M \wedge M \wedge H \wedge L$	{0,0,0,0,0,1}	1
3	$L \wedge L \wedge L \wedge H$	{1,0,0,0,0,0}	1	44	$M \wedge M \wedge H \wedge M$	{0.6,0,0,0,0,0.4}	1
4	$L \wedge L \wedge M \wedge L$	{1,0,0,0,0,0}	1	45	$M \wedge M \wedge H \wedge H$	{0,0.85,0,0,0,0.15}	1
5	$L \wedge L \wedge M \wedge M$	{1,0,0,0,0,0}	1	46	$M \wedge H \wedge L \wedge L$	{0,0.05,0,0,0,0.95}	1
6	$L \wedge L \wedge M \wedge H$	{1,0,0,0,0,0}	1	47	$M \wedge H \wedge L \wedge M$	{0,0.2,0,0,0,0.8}	1
7	$L \wedge L \wedge H \wedge L$	{1,0,0,0,0,0}	1	48	$M \wedge H \wedge L \wedge H$	{0,1,0,0,0,0}	1
8	$L \wedge L \wedge H \wedge M$	{1,0,0,0,0,0}	1	49	$M \wedge H \wedge M \wedge L$	{0,0,0,0,0,1}	1
9	$L \wedge L \wedge H \wedge H$	{1,0,0,0,0,0}	1	50	$M \wedge H \wedge M \wedge M$	{0,0.1,0,0,0,0.9}	1
10	$L \wedge M \wedge L \wedge L$	{0,0.2,0,0,0,0.8}	1	51	$M \wedge H \wedge M \wedge H$	{0,0.15,0,0,0,0.85}	1
11	$L \wedge M \wedge L \wedge M$	{0,0.3,0,0,0,0.7}	1	52	$M \wedge H \wedge H \wedge L$	{0,0,0,0,0,1}	1
12	$L \wedge M \wedge L \wedge H$	{0,1,0,0,0,0}	1	53	$M \wedge H \wedge H \wedge M$	{0,0,0,0,0,1}	1
13	$L \wedge M \wedge M \wedge L$	{0,0,0,0,0,1}	1	54	$M \wedge H \wedge H \wedge H$	{0.1,0,0,0,0,0.9}	1
14	$L \wedge M \wedge M \wedge M$	{0,0.1,0,0,0,0.9}	1	55	$H \wedge L \wedge L \wedge L$	{0,0,0,0,1,0}	1
15	$L \wedge M \wedge M \wedge H$	{0,0.9,0,0,0,0.1}	1	56	$H \wedge L \wedge L \wedge M$	{0,0,0,0,0.9,0.1}	1
16	$L \wedge M \wedge H \wedge L$	{0,0,0,0,0,1}	1	57	$H \wedge L \wedge L \wedge H$	{0,0,0,0,0,1}	1
17	$L \wedge M \wedge H \wedge M$	{0,0,0,0,0,1}	1	58	$H \wedge L \wedge M \wedge L$	{0,0,0,0,0.9,0.1}	1
18	$L \wedge M \wedge H \wedge H$	{0,0.05,0,0,0,0.95}	1	59	$H \wedge L \wedge M \wedge M$	{0,0,0,0,0.8,0.2}	1
19	$L \wedge H \wedge L \wedge L$	{0,0,0,0,0,1}	1	60	$H \wedge L \wedge M \wedge H$	{0,0,0.7,0,0,0.3}	1
20	$L \wedge H \wedge L \wedge M$	{0,0.3,0,0,0,0.7}	1	61	$H \wedge L \wedge H \wedge L$	{0,0,0,0,0,1}	1
21	$L \wedge H \wedge L \wedge H$	{0,0.9,0,0,0,0.1}	1	62	$H \wedge L \wedge H \wedge M$	{0,0,0.7,0,0,0.3}	1
22	$L \wedge H \wedge M \wedge L$	{0,0,0,0,0,1}	1	63	$H \wedge L \wedge H \wedge H$	{0,0,1,0,0,0}	1
23	$L \wedge H \wedge M \wedge M$	{0,0.4,0,0,0,0.6}	1	64	$H \wedge M \wedge L \wedge L$	{0,0,0,0,0.05,0.95}	1
24	$L \wedge H \wedge M \wedge H$	{0,0.8,0,0,0,0.2}	1	65	$H \wedge M \wedge L \wedge M$	{0,0,0,0,0,1}	1
25	$L \wedge H \wedge H \wedge L$	{0,0,0,0,0,1}	1	66	$H \wedge M \wedge L \wedge H$	{0,0,0,0,0,1}	1
26	$L \wedge H \wedge H \wedge M$	{0,0,0,0,0,1}	1	67	$H \wedge M \wedge M \wedge L$	{0,0,0,0,0.05,0.95}	1
27	$L \wedge H \wedge H \wedge H$	{0,0,0,0,0,1}	1	68	$H \wedge M \wedge M \wedge M$	{0,0,0,0,0.03,0.97}	1
28	$M \wedge L \wedge L \wedge L$	{0,0,0,1,0,0}	1	69	$H \wedge M \wedge M \wedge H$	{0,0,0,0,0,1}	1
29	$M \wedge L \wedge L \wedge M$	{0,0,0,0.8,0,0.2}	1	70	$H \wedge M \wedge H \wedge L$	{0,0,0,0,0,1}	1
30	$M \wedge L \wedge L \wedge H$	{0.1,0,0,0,0,0.9}	1	71	$H \wedge M \wedge H \wedge M$	{0,0,0.2,0,0,0.8}	1
31	$M \wedge L \wedge M \wedge L$	{0,0,0.8,0,0,0.2}	1	72	$H \wedge M \wedge H \wedge H$	{0,0,0.3,0,0,0.7}	1
32	$M \wedge L \wedge M \wedge M$	{0.3,0,0,0.5,0,0.1}	1	73	$H \wedge H \wedge L \wedge L$	{0,0,0,0,0,1}	1
33	$M \wedge L \wedge M \wedge H$	{0.6,0,0,0,0,0.4}	1	74	$H \wedge H \wedge L \wedge M$	{0,0,0,0,0,1}	1
34	$M \wedge L \wedge H \wedge L$	{0.1,0,0,0,0,0.9}	1	75	$H \wedge H \wedge L \wedge H$	{0,0,0,0,0,1}	1
35	$M \wedge L \wedge H \wedge M$	{0.6,0,0,0,0,0.4}	1	76	$H \wedge H \wedge M \wedge L$	{0,0,0,0,0,1}	1
36	$M \wedge L \wedge H \wedge H$	{0.8,0,0,0,0,0.2}	1	77	$H \wedge H \wedge M \wedge M$	{0,0,0,0,0,1}	1
37	$M \wedge M \wedge L \wedge L$	{0,0,0,0,0,1}	1	78	$H \wedge H \wedge M \wedge H$	{0,0,0,0,0,1}	1
38	$M \wedge M \wedge L \wedge M$	{0,0.15,0,0,0,0.85}	1	79	$H \wedge H \wedge H \wedge L$	{0,0,0,0,0,1}	1
39	$M \wedge M \wedge L \wedge H$	{0,0.2,0,0,0,0.8}	1	80	$H \wedge H \wedge H \wedge M$	{0,0,0,0,0,1}	1
40	$M \wedge M \wedge M \wedge L$	{0,0,0,0,0,1}	1	81	$H \wedge H \wedge H \wedge H$	{0,0,0,0,0,1}	1
41	$M \wedge M \wedge M \wedge M$	{1,0,0,0,0,0}	1				

One of the future works is how to process data to improve the accuracy of fault diagnosis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A.

See Table A1 and Table A2.

Appendix B.

See Table B1, Table B2 and Table B3.

Appendix C.

See Table C1 and Table C2.

Appendix D.

See Table D1 and Table D2.

Table D2 Rule base of BRB1.

No.	$x_1 \wedge x_2 \wedge x_3 \wedge x_4$	$\{D_1, D_2, D_3, D_4, D_5, D_6\}$	θ_k	No.	$x_1 \wedge x_2 \wedge x_3 \wedge x_4$	$\{D_1, D_2, D_3, D_4, D_5, D_6\}$	θ_k
1	$L \wedge L \wedge L \wedge L$	{0.11,0.06,0.07,0.28,0.24,0.24}	0.26	42	$M \wedge M \wedge M \wedge H$	{0.24,0.37,0.17,0.14,0.03,0.05}	0.81
2	$L \wedge L \wedge L \wedge M$	{0.09,0.06,0.11,0.16,0.43,0.15}	0.8	43	$M \wedge M \wedge H \wedge L$	{0.22,0.13,0.3,0.22,0.04,0.09}	0.05
3	$L \wedge L \wedge L \wedge H$	{0.05,0.24,0.18,0.18,0.18,0.17}	0.34	44	$M \wedge M \wedge H \wedge M$	{0.19,0.1,0.44,0.12,0.09,0.06}	0.45
4	$L \wedge L \wedge M \wedge L$	{0.12,0.07,0.08,0.46,0.14,0.13}	0.15	45	$M \wedge M \wedge H \wedge H$	{0.08,0.28,0.48,0.1,0.02,0.04}	0.5
5	$L \wedge L \wedge M \wedge M$	{0.12,0.07,0.15,0.29,0.28,0.09}	0.75	46	$M \wedge H \wedge L \wedge L$	{0.17,0.1,0.07,0.15,0.07,0.44}	0.74
6	$L \wedge L \wedge M \wedge H$	{0.06,0.24,0.2,0.3,0.1,0.1}	0.83	47	$M \wedge H \wedge L \wedge M$	{0.18,0.1,0.13,0.1,0.15,0.34}	0.18
7	$L \wedge L \wedge H \wedge L$	{0.04,0.06,0.28,0.38,0.13,0.11}	0.54	48	$M \wedge H \wedge L \wedge H$	{0.09,0.31,0.16,0.09,0.04,0.31}	0.28
8	$L \wedge L \wedge H \wedge M$	{0.04,0.05,0.41,0.21,0.22,0.07}	0.88	49	$M \wedge H \wedge M \wedge L$	{0.21,0.12,0.09,0.27,0.04,0.27}	0.33
9	$L \wedge L \wedge H \wedge H$	{0.01,0.16,0.5,0.19,0.07,0.07}	0.46	50	$M \wedge H \wedge M \wedge M$	{0.22,0.12,0.16,0.19,0.09,0.22}	0.33
10	$L \wedge M \wedge L \wedge L$	{0.28,0.18,0.06,0.11,0.09,0.28}	0.05	51	$M \wedge H \wedge M \wedge H$	{0.1,0.34,0.19,0.16,0.02,0.19}	0.39
11	$L \wedge M \wedge L \wedge M$	{0.28,0.17,0.1,0.07,0.18,0.2}	0.93	52	$M \wedge H \wedge H \wedge L$	{0.08,0.1,0.3,0.23,0.03,0.26}	0.78
12	$L \wedge M \wedge L \wedge H$	{0.13,0.46,0.12,0.06,0.05,0.18}	0.89	53	$M \wedge H \wedge H \wedge M$	{0.08,0.09,0.46,0.13,0.07,0.17}	0.59
13	$L \wedge M \wedge M \wedge L$	{0.32,0.2,0.07,0.2,0.05,0.16}	0.54	54	$M \wedge H \wedge H \wedge H$	{0.03,0.24,0.49,0.1,0.01,0.13}	0.59
14	$L \wedge M \wedge M \wedge M$	{0.33,0.18,0.13,0.13,0.11,0.12}	0.81	55	$H \wedge L \wedge L \wedge L$	{0.12,0.06,0.07,0.17,0.44,0.14}	0.45
15	$L \wedge M \wedge M \wedge H$	{0.15,0.48,0.15,0.1,0.03,0.09}	0.62	56	$H \wedge L \wedge L \wedge M$	{0.09,0.04,0.09,0.08,0.63,0.07}	0.96
16	$L \wedge M \wedge H \wedge L$	{0.15,0.18,0.28,0.19,0.05,0.15}	0.95	57	$H \wedge L \wedge L \wedge H$	{0.07,0.2,0.18,0.1,0.35,0.1}	0.91
17	$L \wedge M \wedge H \wedge M$	{0.13,0.16,0.41,0.1,0.1,0.1}	0.77	58	$H \wedge L \wedge M \wedge L$	{0.15,0.07,0.1,0.31,0.28,0.09}	0.52
18	$L \wedge M \wedge H \wedge H$	{0.05,0.37,0.41,0.07,0.02,0.08}	0.52	59	$H \wedge L \wedge M \wedge M$	{0.13,0.05,0.14,0.17,0.47,0.04}	0.8
19	$L \wedge H \wedge L \wedge L$	{0.09,0.12,0.05,0.1,0.06,0.58}	0.53	60	$H \wedge L \wedge M \wedge H$	{0.08,0.22,0.23,0.2,0.21,0.06}	0.34
20	$L \wedge H \wedge L \wedge M$	{0.1,0.13,0.1,0.07,0.14,0.46}	0.75	61	$H \wedge L \wedge H \wedge L$	{0.06,0.05,0.31,0.25,0.26,0.07}	0.04
21	$L \wedge H \wedge L \wedge H$	{0.05,0.36,0.1,0.06,0.04,0.39}	0.44	62	$H \wedge L \wedge H \wedge M$	{0.04,0.04,0.39,0.12,0.38,0.03}	0.46
22	$L \wedge H \wedge M \wedge L$	{0.13,0.15,0.07,0.21,0.04,0.4}	0.31	63	$H \wedge L \wedge H \wedge H$	{0.02,0.14,0.54,0.12,0.15,0.03}	0.82
23	$L \wedge H \wedge M \wedge M$	{0.14,0.16,0.14,0.14,0.1,0.32}	0.2	64	$H \wedge M \wedge L \wedge L$	{0.34,0.16,0.07,0.06,0.19,0.18}	0.35
24	$L \wedge H \wedge M \wedge H$	{0.06,0.41,0.15,0.11,0.02,0.25}	0.8	65	$H \wedge M \wedge L \wedge M$	{0.3,0.13,0.1,0.03,0.33,0.11}	0.83
25	$L \wedge H \wedge H \wedge L$	{0.05,0.14,0.25,0.17,0.04,0.35}	0.57	66	$H \wedge M \wedge L \wedge H$	{0.17,0.42,0.14,0.03,0.12,0.12}	0.54
26	$L \wedge H \wedge H \wedge M$	{0.05,0.12,0.4,0.1,0.08,0.25}	0.89	67	$H \wedge M \wedge M \wedge L$	{0.4,0.17,0.08,0.13,0.12,0.1}	0.04
27	$L \wedge H \wedge H \wedge H$	{0.01,0.3,0.41,0.08,0.01,0.19}	0.45	68	$H \wedge M \wedge M \wedge M$	{0.37,0.15,0.13,0.07,0.21,0.07}	0.2
28	$M \wedge L \wedge L \wedge L$	{0.16,0.04,0.08,0.34,0.23,0.15}	0.33	69	$H \wedge M \wedge M \wedge H$	{0.19,0.44,0.17,0.07,0.07,0.06}	0.72
29	$M \wedge L \wedge L \wedge M$	{0.14,0.04,0.13,0.19,0.4,0.1}	0.1	70	$H \wedge M \wedge H \wedge L$	{0.19,0.16,0.31,0.12,0.12,0.1}	0.97
30	$M \wedge L \wedge L \wedge H$	{0.09,0.18,0.21,0.23,0.18,0.11}	0.56	71	$H \wedge M \wedge H \wedge M$	{0.15,0.12,0.42,0.06,0.19,0.06}	0.64
31	$M \wedge L \wedge M \wedge L$	{0.16,0.05,0.09,0.51,0.12,0.07}	0.97	72	$H \wedge M \wedge H \wedge H$	{0.07,0.33,0.46,0.05,0.05,0.04}	0.68
32	$M \wedge L \wedge M \wedge M$	{0.17,0.04,0.15,0.34,0.24,0.06}	0	73	$H \wedge H \wedge L \wedge L$	{0.14,0.13,0.07,0.07,0.15,0.44}	0.12
33	$M \wedge L \wedge M \wedge H$	{0.1,0.17,0.23,0.36,0.09,0.05}	0.43	74	$H \wedge H \wedge L \wedge M$	{0.13,0.11,0.11,0.04,0.29,0.32}	0.87
34	$M \wedge L \wedge H \wedge L$	{0.07,0.04,0.29,0.43,0.11,0.06}	0.27	75	$H \wedge H \wedge L \wedge H$	{0.07,0.36,0.15,0.04,0.1,0.28}	0.69
35	$M \wedge L \wedge H \wedge M$	{0.06,0.03,0.43,0.24,0.2,0.04}	0.87	76	$H \wedge H \wedge M \wedge L$	{0.19,0.16,0.09,0.16,0.1,0.3}	0.38
36	$M \wedge L \wedge H \wedge H$	{0.03,0.11,0.54,0.22,0.06,0.04}	0.72	77	$H \wedge H \wedge M \wedge M$	{0.18,0.14,0.16,0.09,0.21,0.22}	0.33
37	$M \wedge M \wedge L \wedge L$	{0.41,0.13,0.07,0.14,0.09,0.16}	0.2	78	$H \wedge H \wedge M \wedge H$	{0.08,0.41,0.19,0.08,0.06,0.18}	0.88
38	$M \wedge M \wedge L \wedge M$	{0.4,0.12,0.11,0.08,0.17,0.12}	0.07	79	$H \wedge H \wedge H \wedge L$	{0.07,0.13,0.31,0.13,0.09,0.27}	0.77
39	$M \wedge M \wedge L \wedge H$	{0.22,0.37,0.15,0.08,0.06,0.12}	0.17	80	$H \wedge H \wedge H \wedge M$	{0.06,0.11,0.44,0.06,0.16,0.17}	0.47
40	$M \wedge M \wedge M \wedge L$	{0.43,0.13,0.08,0.23,0.04,0.09}	0.26	81	$H \wedge H \wedge H \wedge H$	{0.02,0.28,0.48,0.05,0.04,0.13}	0.84
41	$M \wedge M \wedge M \wedge M$	{0.44,0.12,0.13,0.15,0.1,0.06}	0.78				

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